Population Pyramids & Age Structure:
Does Age Really Matter?

LECTURE 1.2

Sociology 170 – Population Problems

Spring 2026

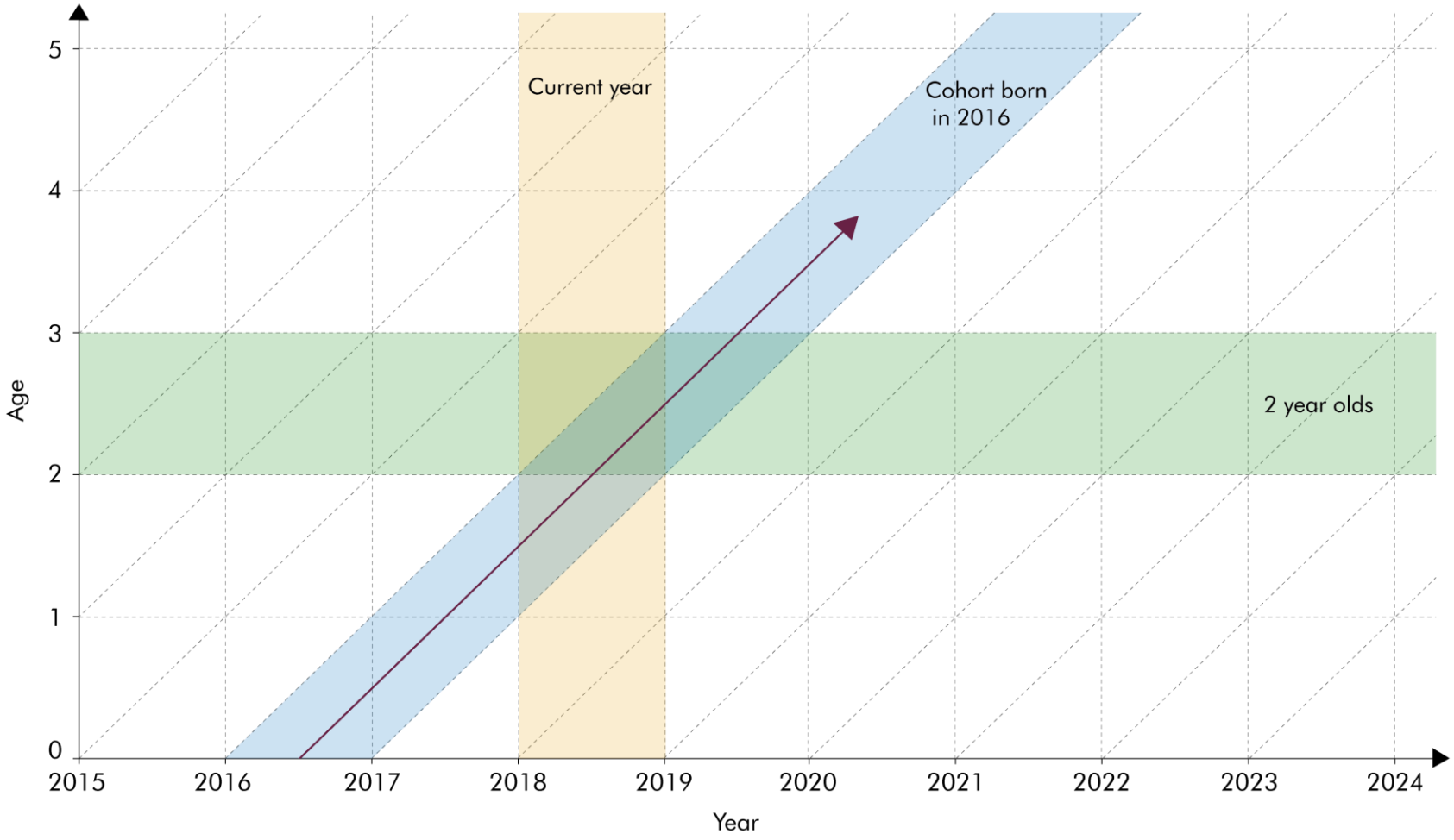
3 Concepts

1. **Age**
2. **Period**: Time (often year)
3. **Cohort**: Group born at the same time

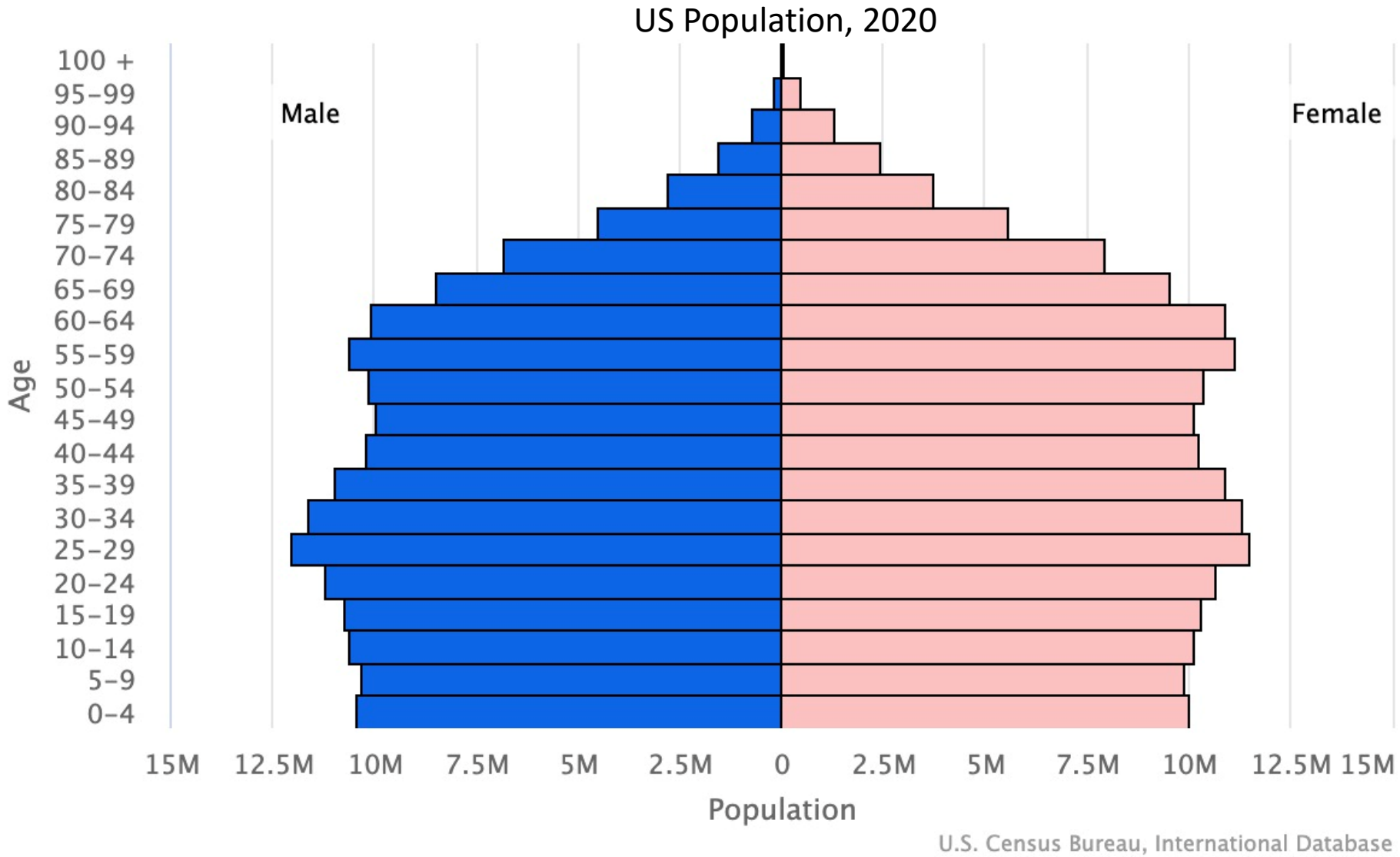
Cohort = Period-Age

Period = Cohort + Age

Age = Period – Cohort



Anatomy of a Population Pyramid



Population Pyramids: Past, Present, and Future

- **Past** – preserves past events in the age structure
  - *e.g. what impact did past events have on populations?*

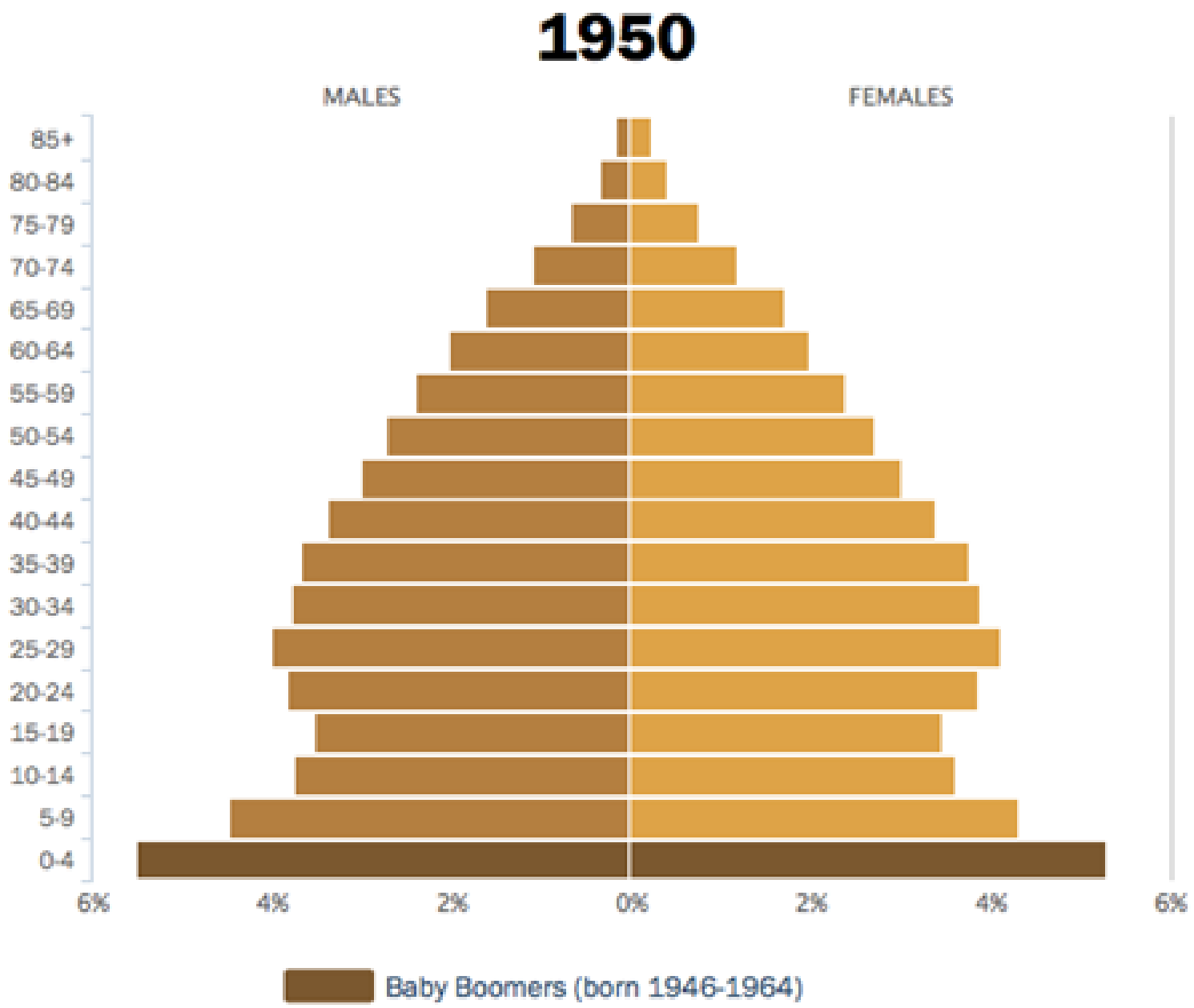
- **Present** – allows us to easily see who is at risk of demographic events in the immediate future.
  - *e.g. what percent of the population is at high risk of mortality?*

- **Future** – permits inferences about future outlook of a population (2 parts – *growth/decline* + *dependency ratios*)
  - *e.g. how many people will need Medicare in 20 years?*

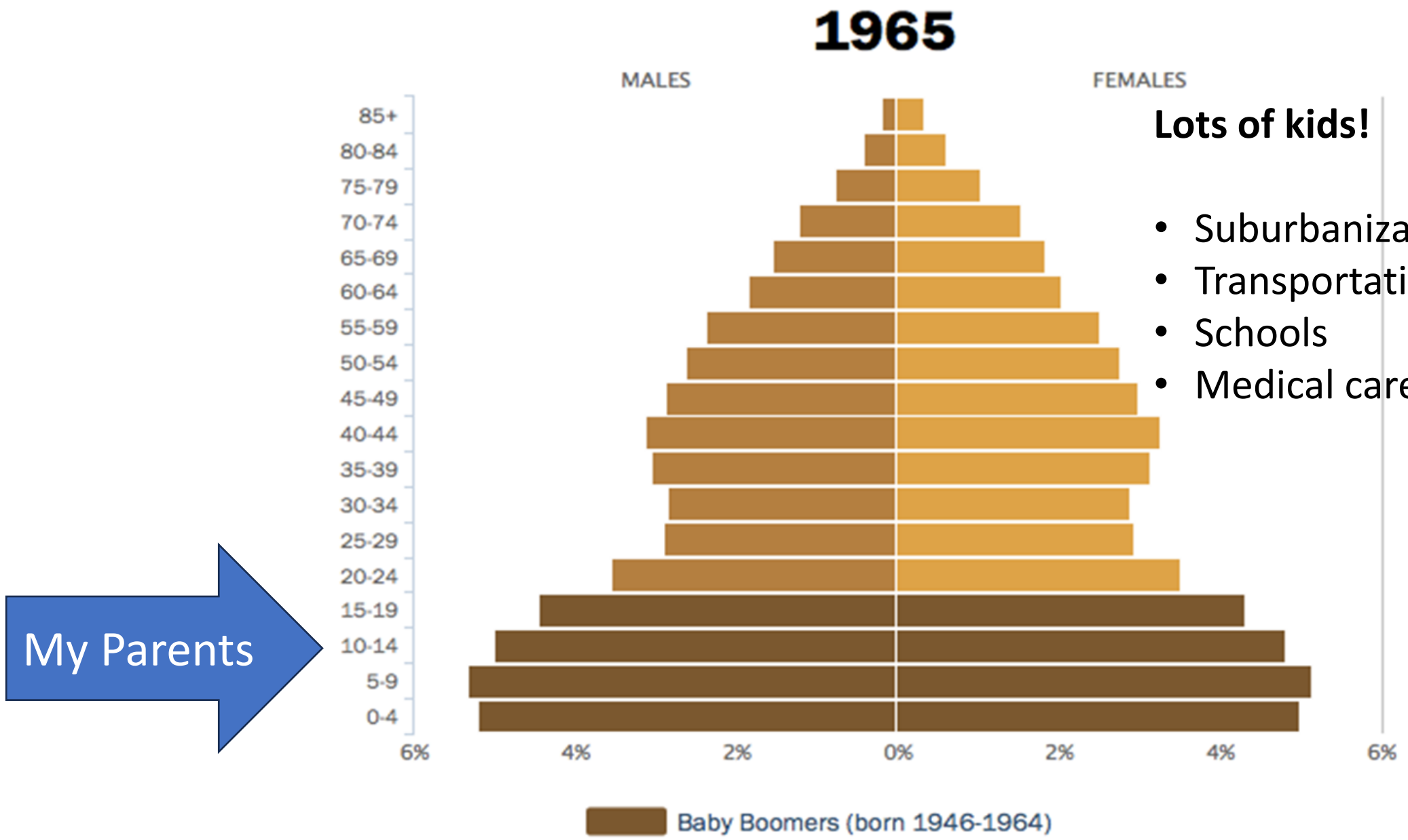
Population Pyramid Example: Baby Boomers

- Can we see their past?
  - (When was the baby boom? Why was it then?)
- At any point in their lives, were they at risk of specific events?
  - (Fertility? Mortality?)
- Could we use population pyramids to predict their future? Or our country's?

U.S. population by age group, 1950-2060



U.S. population by age group, 1950-2060

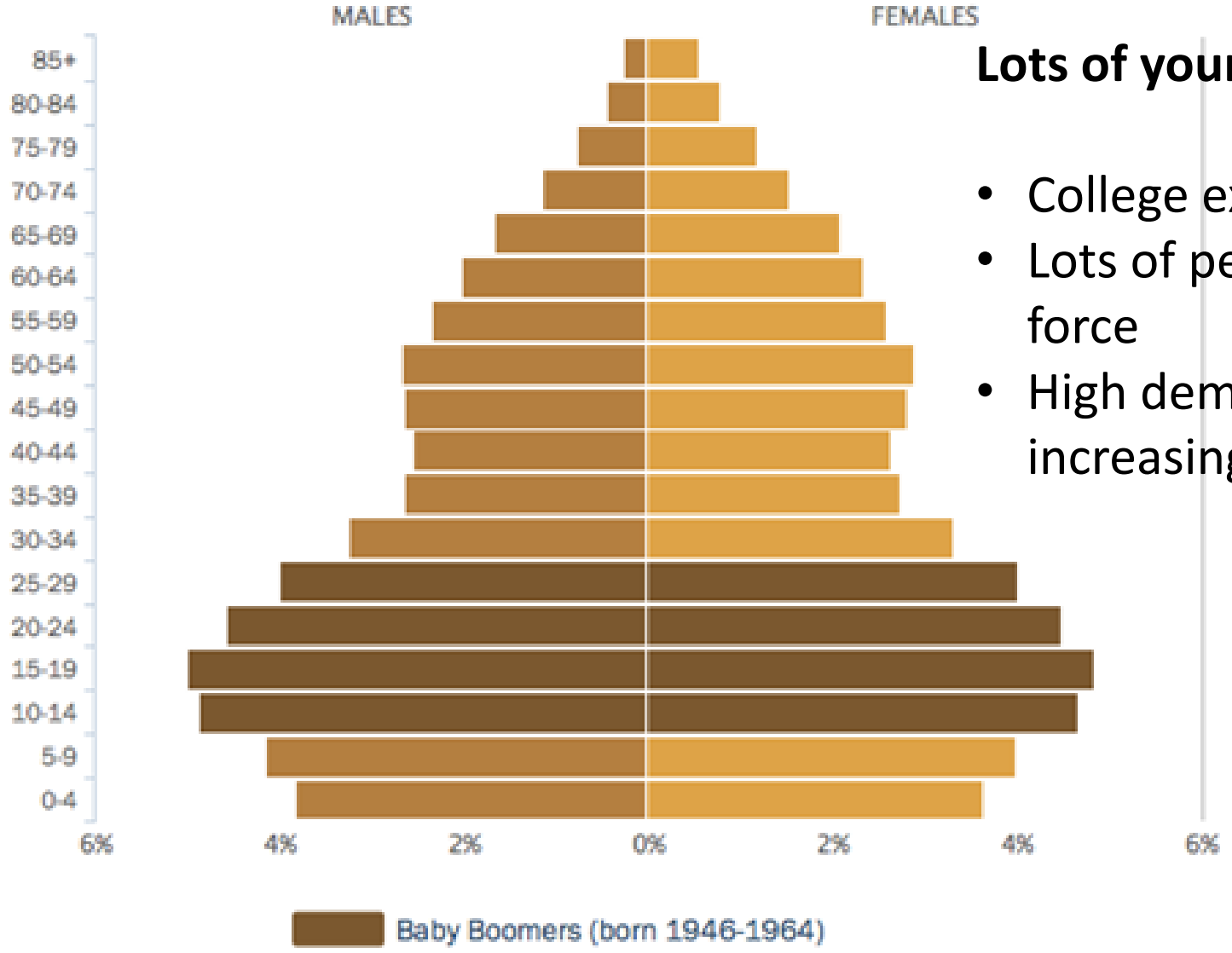


Lots of kids!

- Suburbanization
- Transportation infrastructure
- Schools
- Medical care

U.S. population by age group, 1950-2060

1975

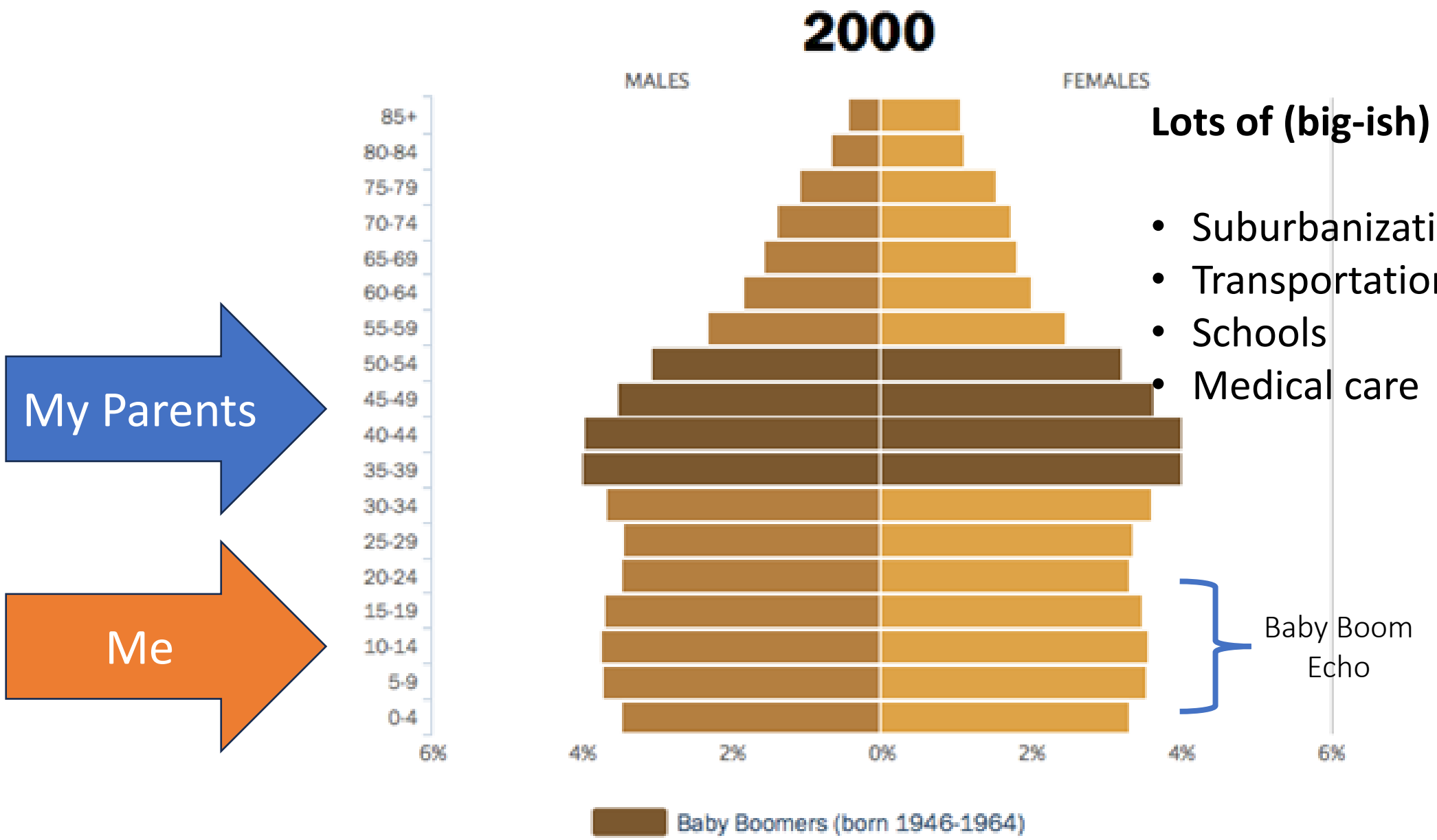


Lots of young adults!

- College expansion
- Lots of people entering labor force
- High demand for housing → increasing suburbanization

My Parents

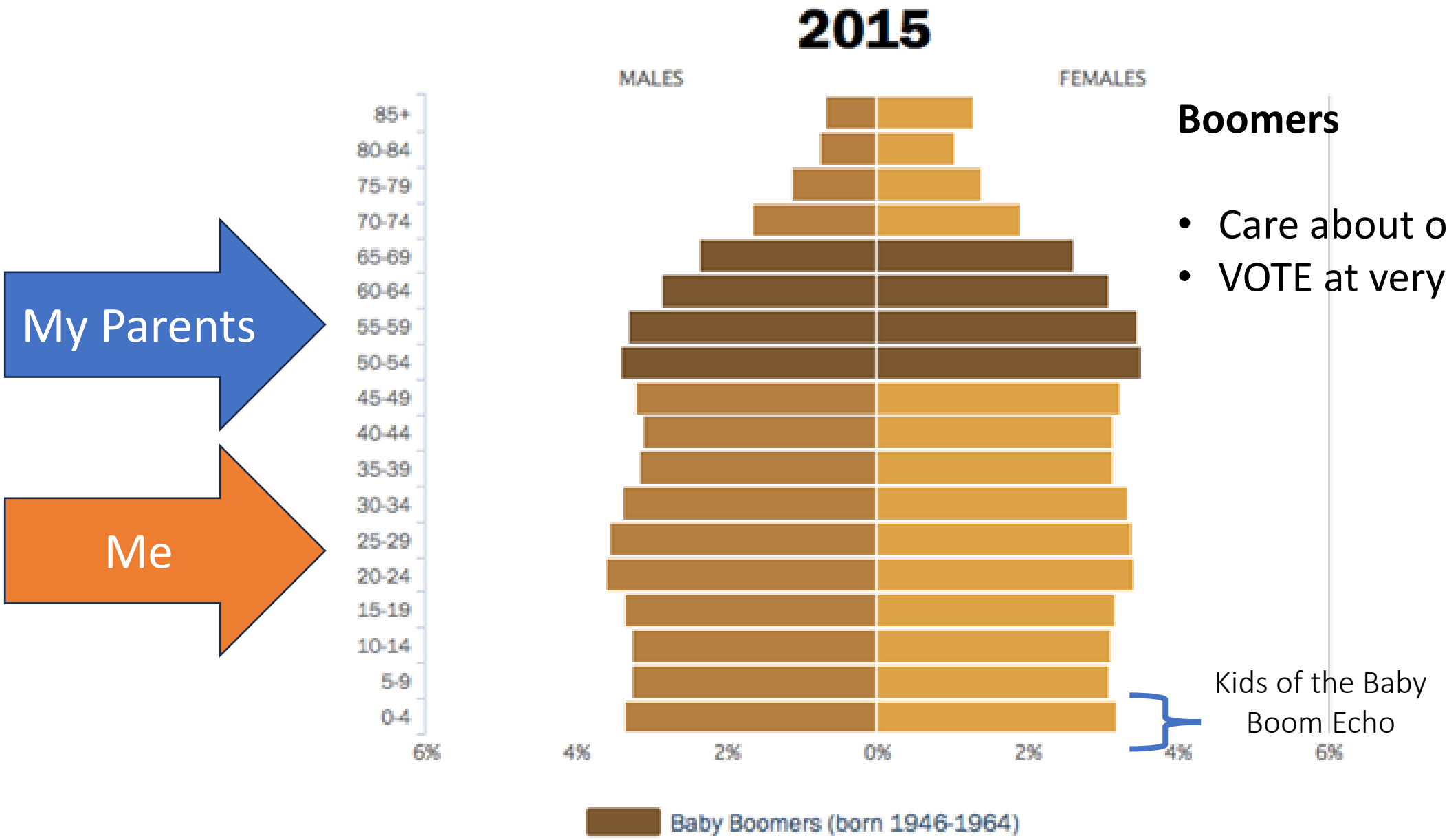
U.S. population by age group, 1950-2060



Lots of (big-ish) families!

- Suburbanization
- Transportation infrastructure
- Schools
- Medical care

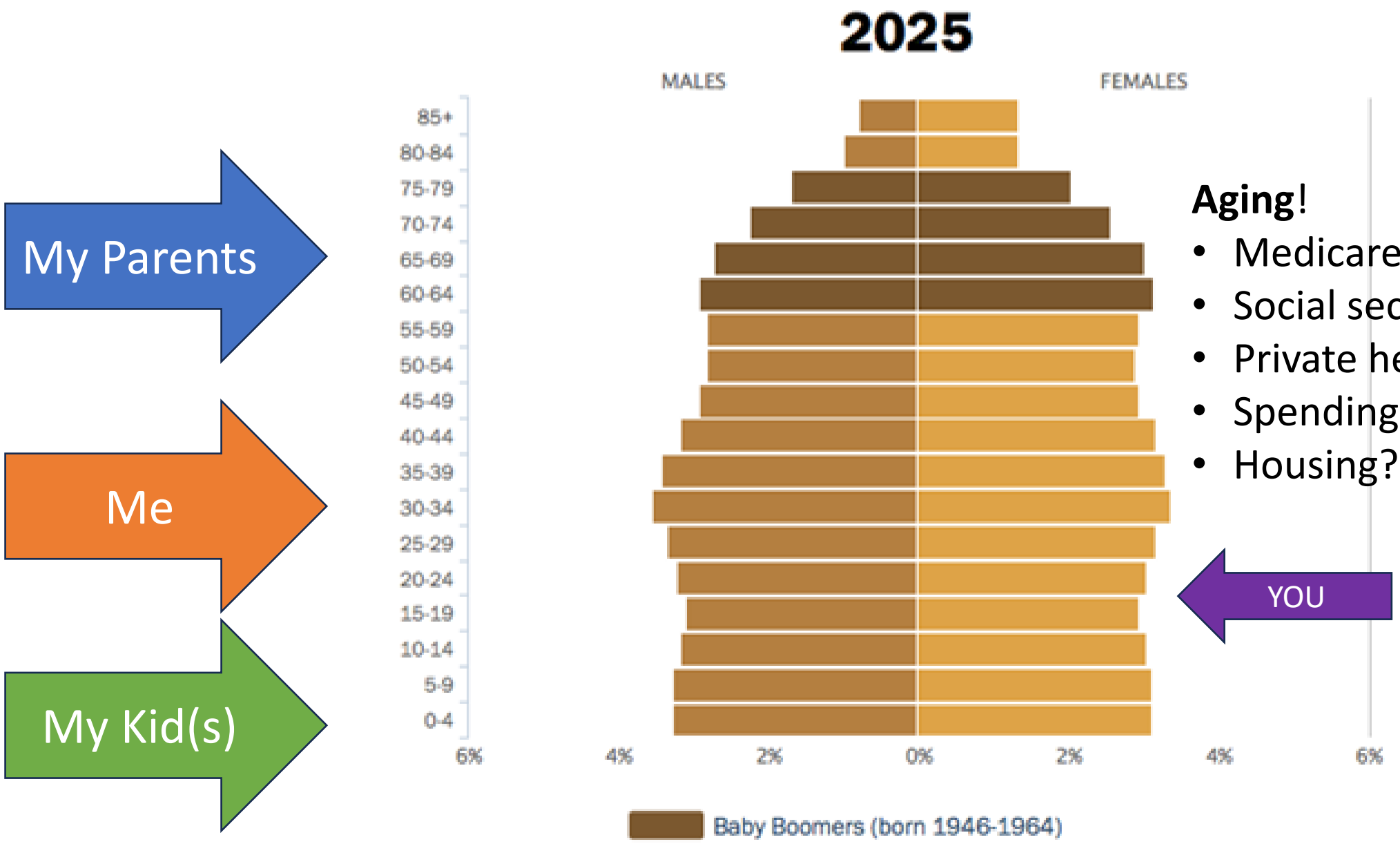
U.S. population by age group, 1950-2060



Boomers

- Care about old people stuff
- VOTE at very high rates

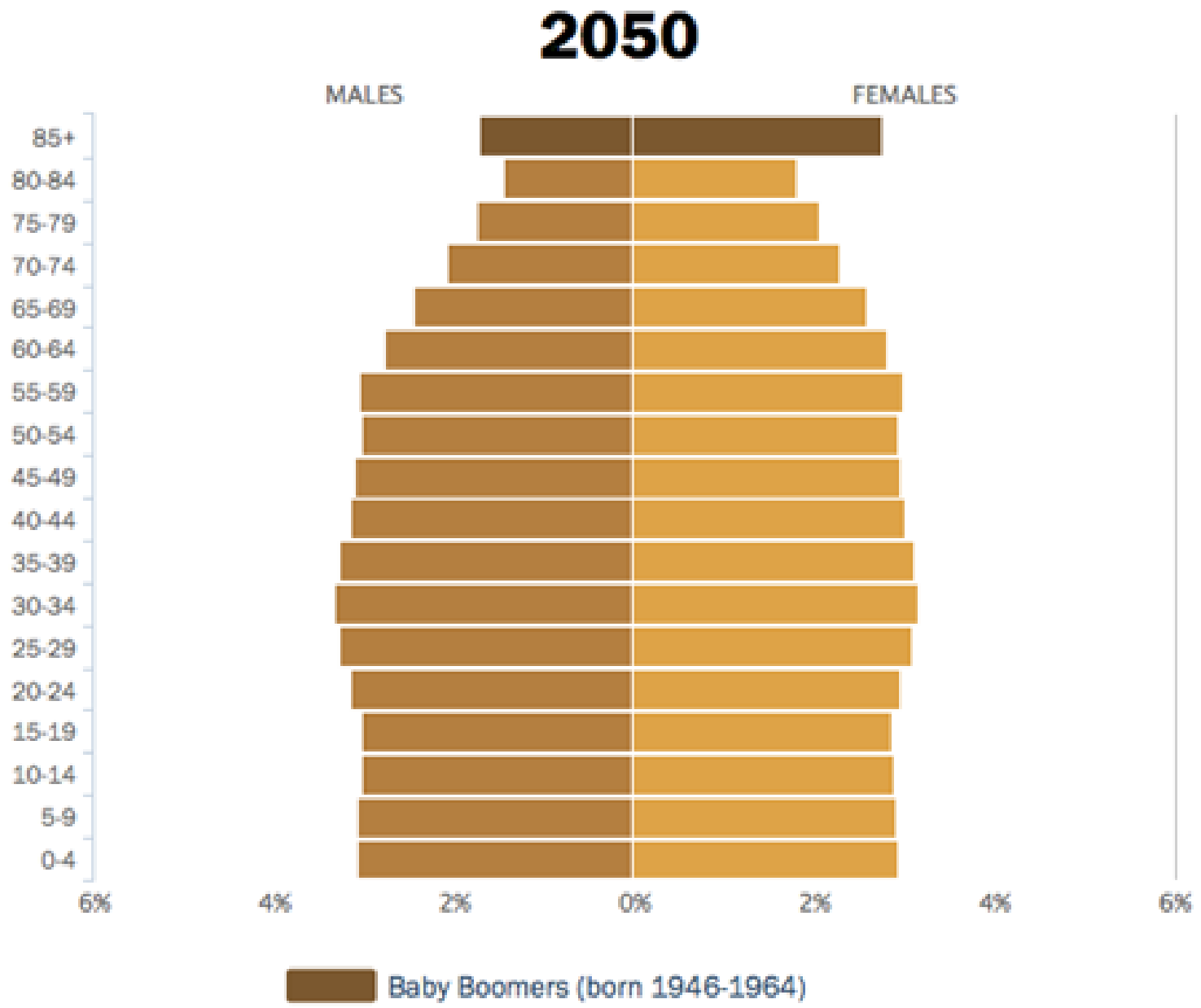
U.S. population by age group, 1950-2060



Aging!

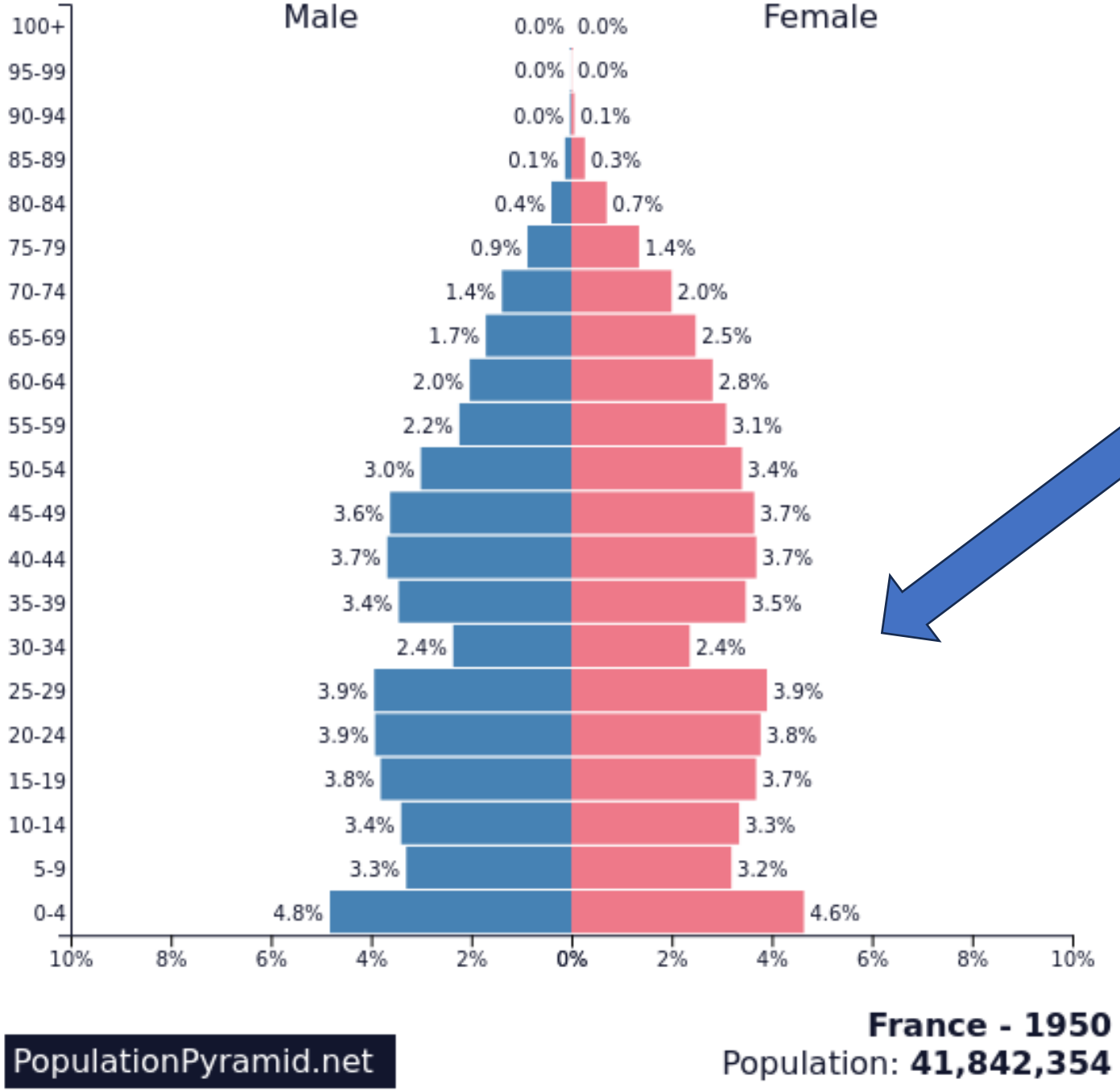
- Medicare
- Social security
- Private healthcare services
- Spending of savings
- Housing?

U.S. population by age group, 1950-2060



4 Reasons Age Structure Matters:

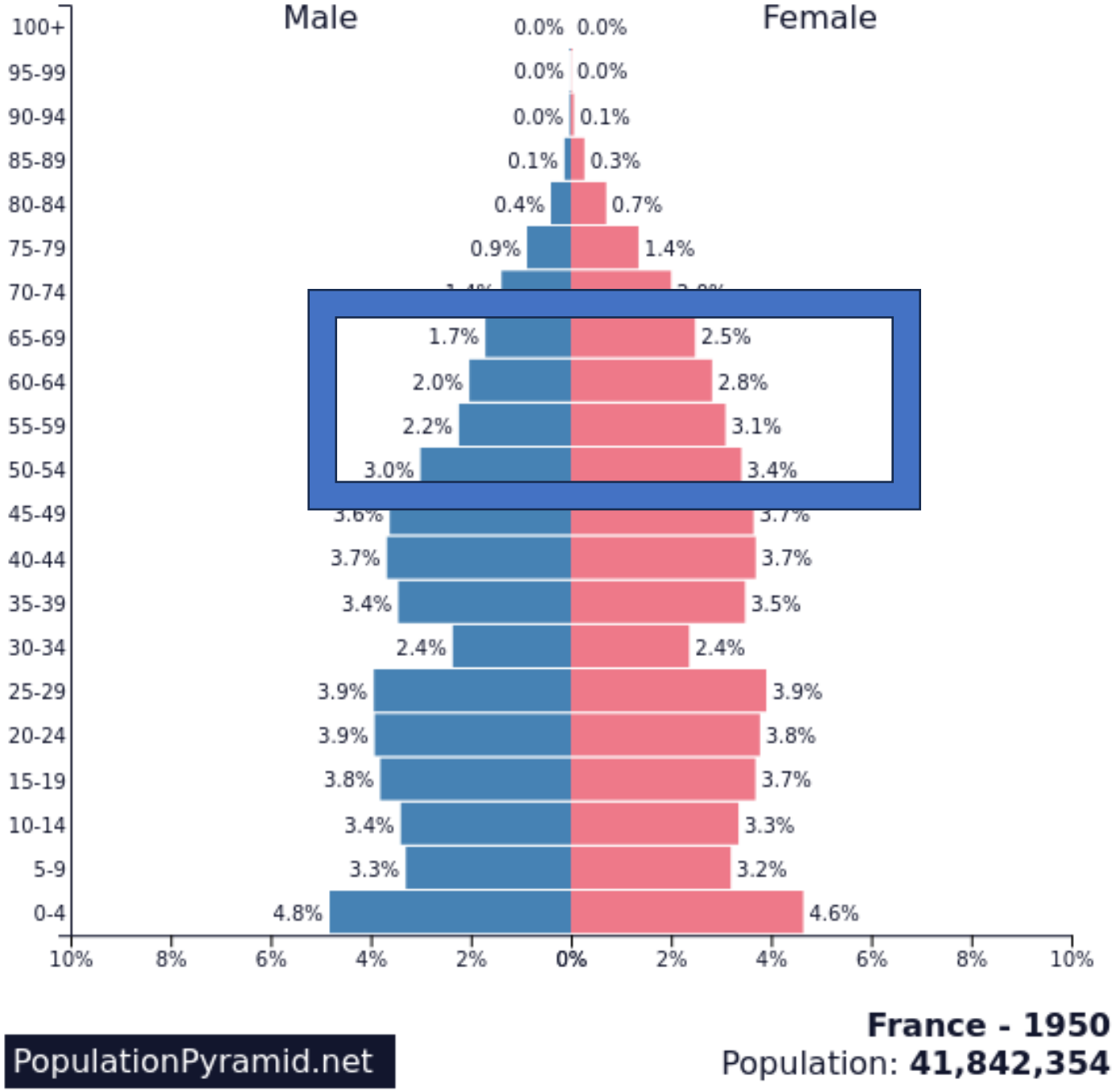
1. Record of past cohort experiences
2. Age pattern of demographic risk
3. Potential for population growth
4. Age dependency ratios



France 1950

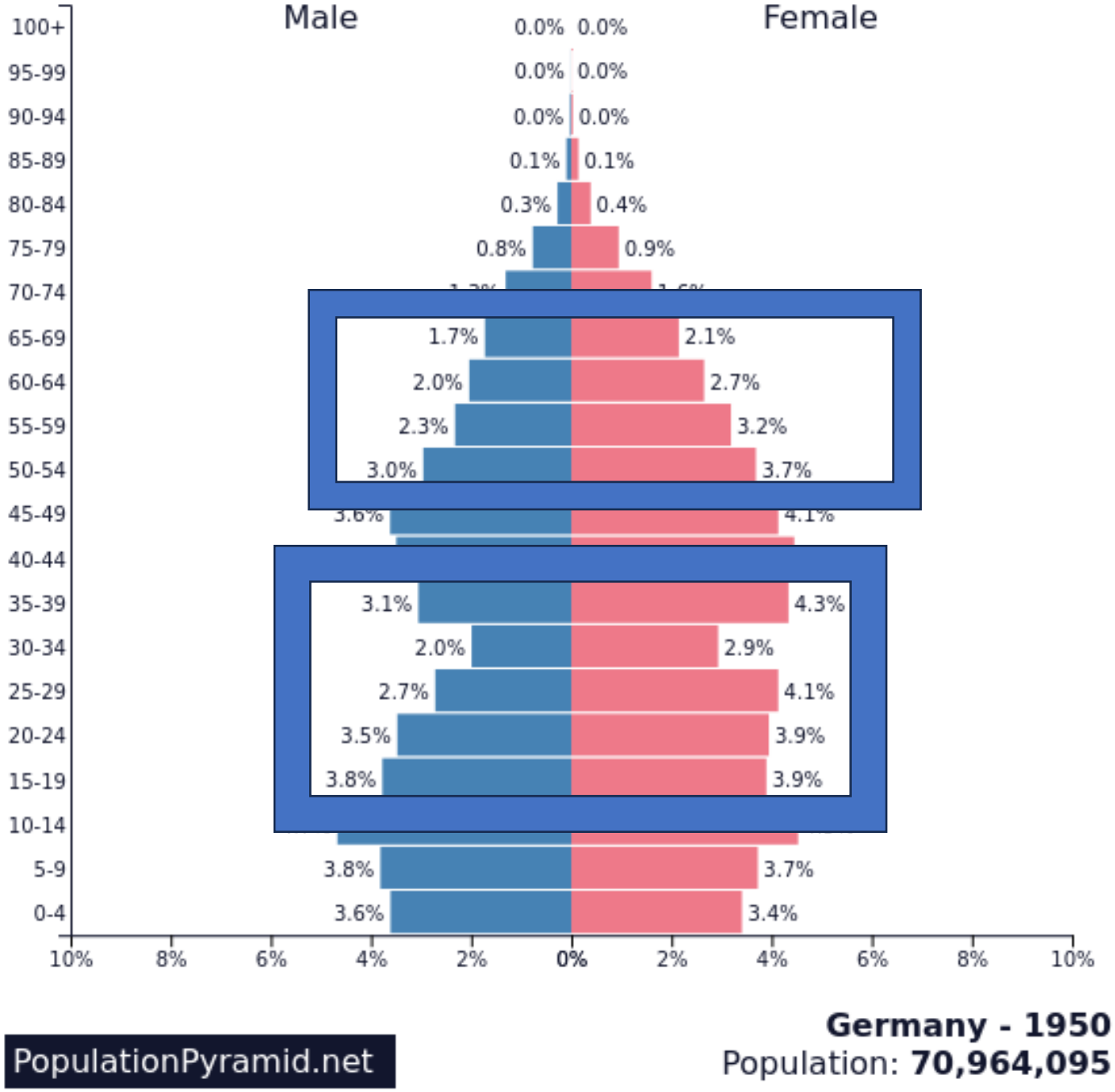
What happened here?

These people were from the 1916-1920 cohort.



France 1950

WWI – many more women than men among cohorts that were 20-30 in 1914-18



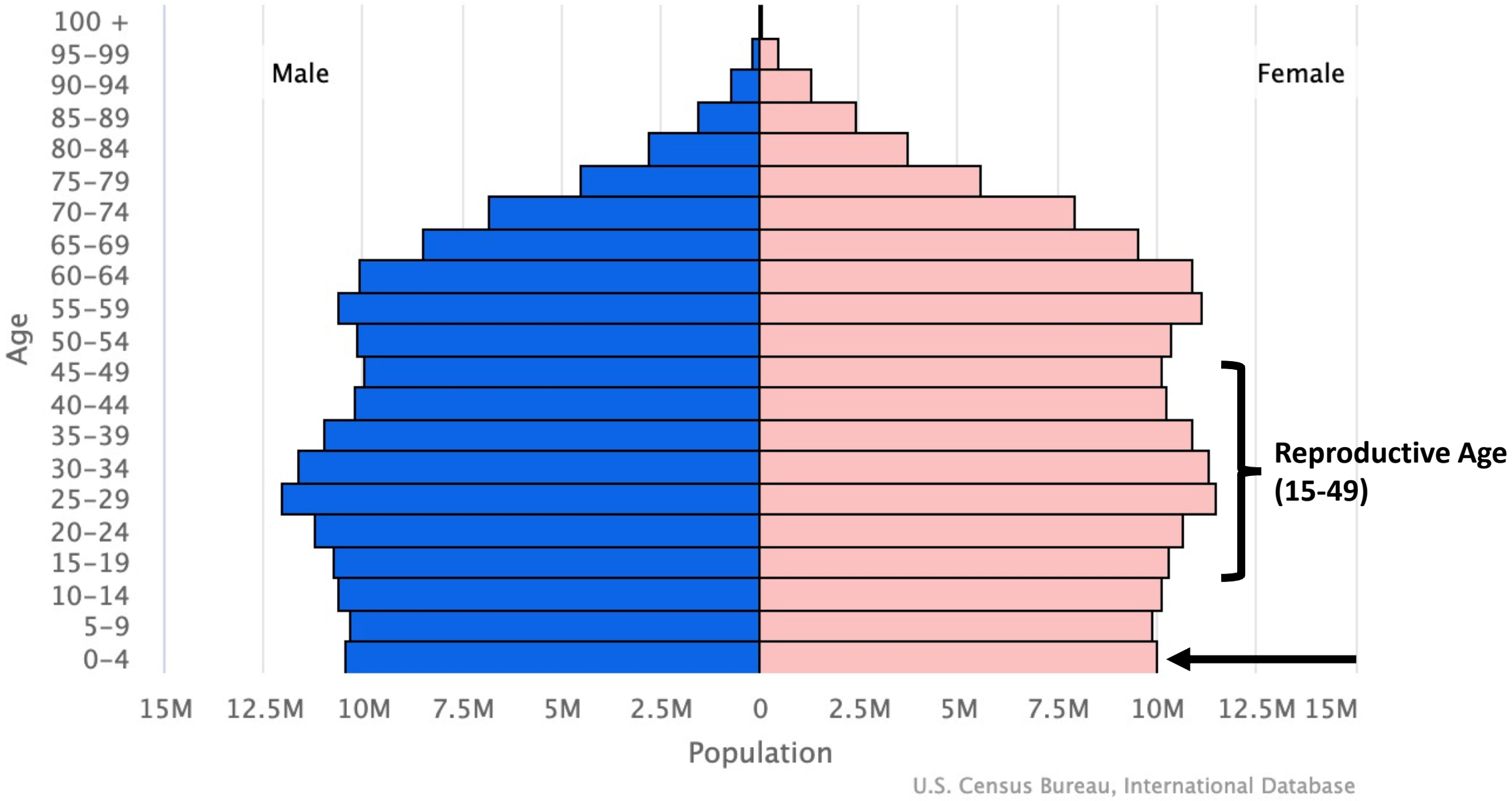
Germany 1950

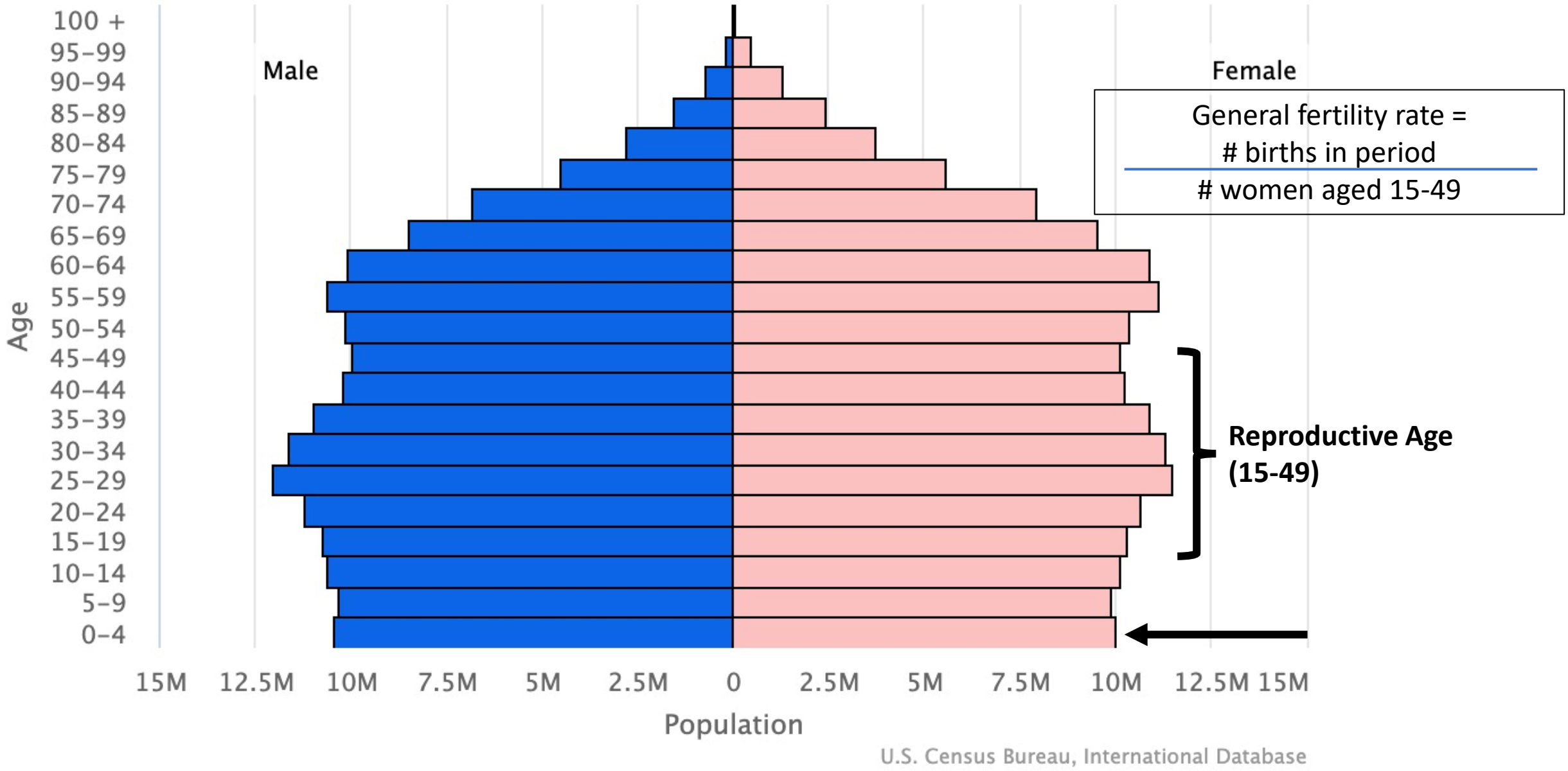
WWI

WWII – Germany lost a lot more men than France

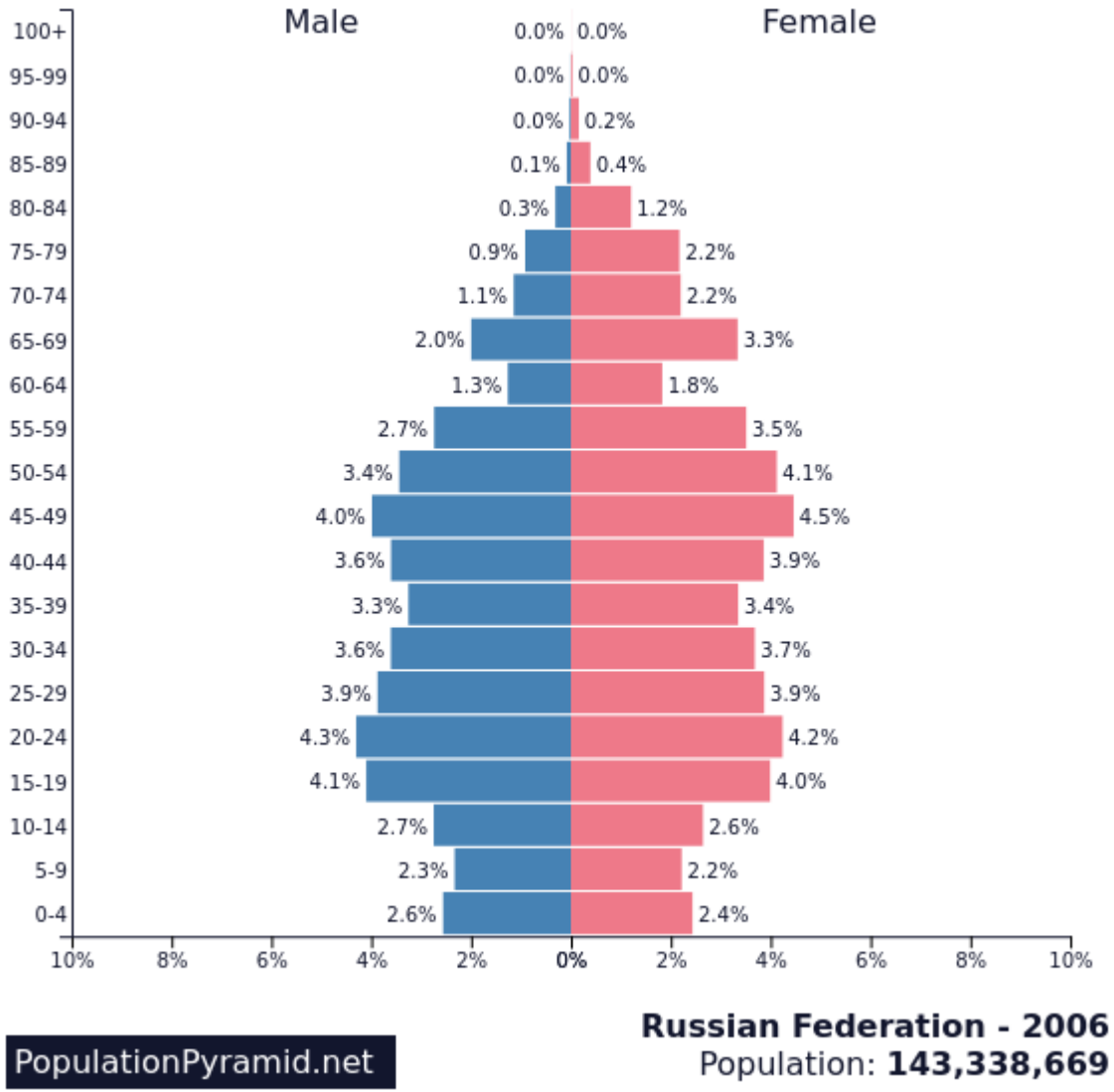
Why does age structure matter?

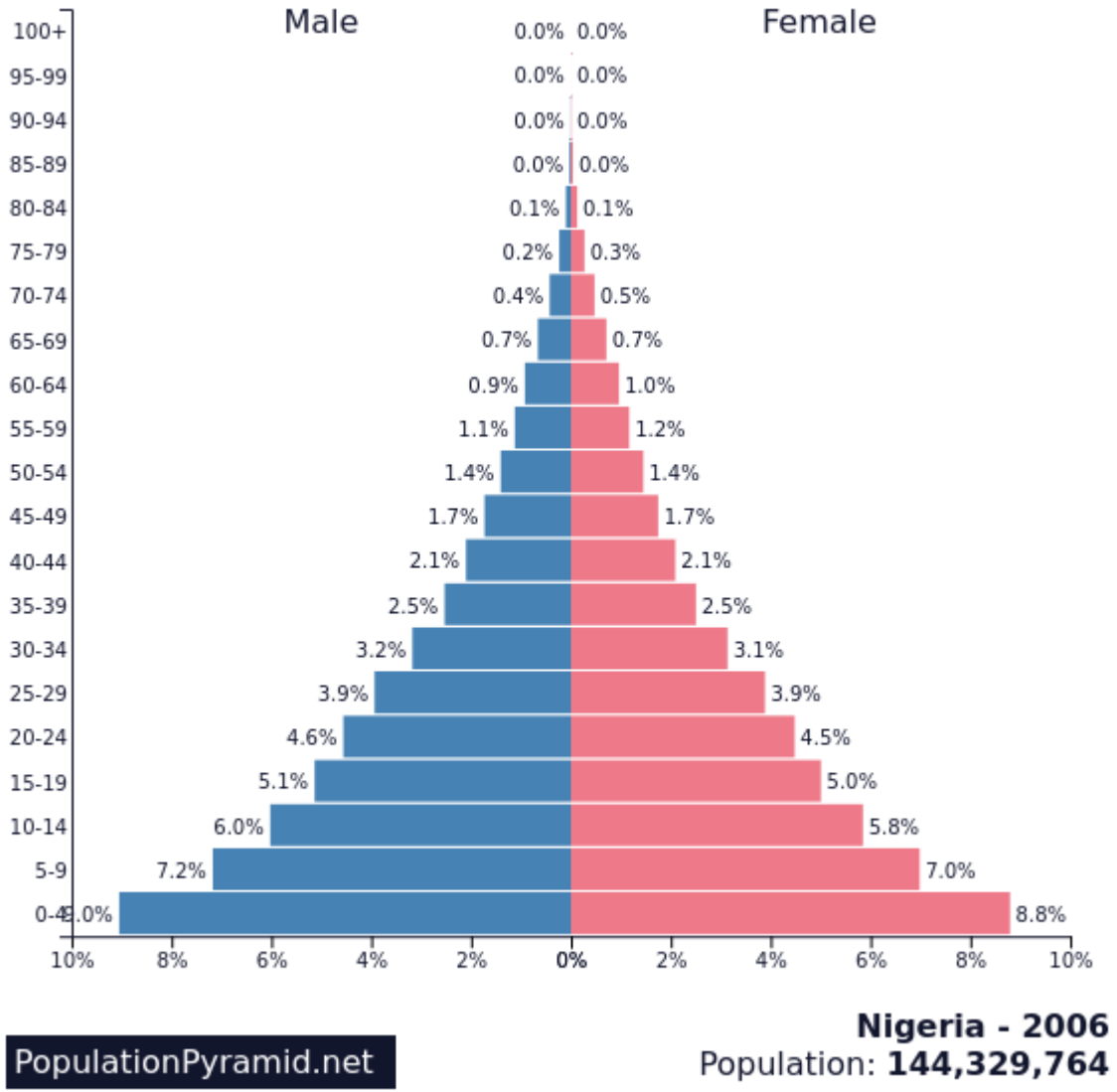
1. Record of past cohort experiences
2. Age pattern of demographic risk
3. Potential for population growth
4. Age dependency ratios


Male
Female
Age
100 +
95-99
90-94
85-89
80-84
75-79
70-74
65-69
60-64
55-59
50-54
45-49
40-44
35-39
30-34
25-29
20-24
15-19
10-14
5-9
0-4
15M
12.5M
10M
7.5M
5M
2.5M
0
2.5M
5M
7.5M
10M
12.5M
15M
Population
Reproductive Age
(15-49)
U.S. Census Bureau, International Database

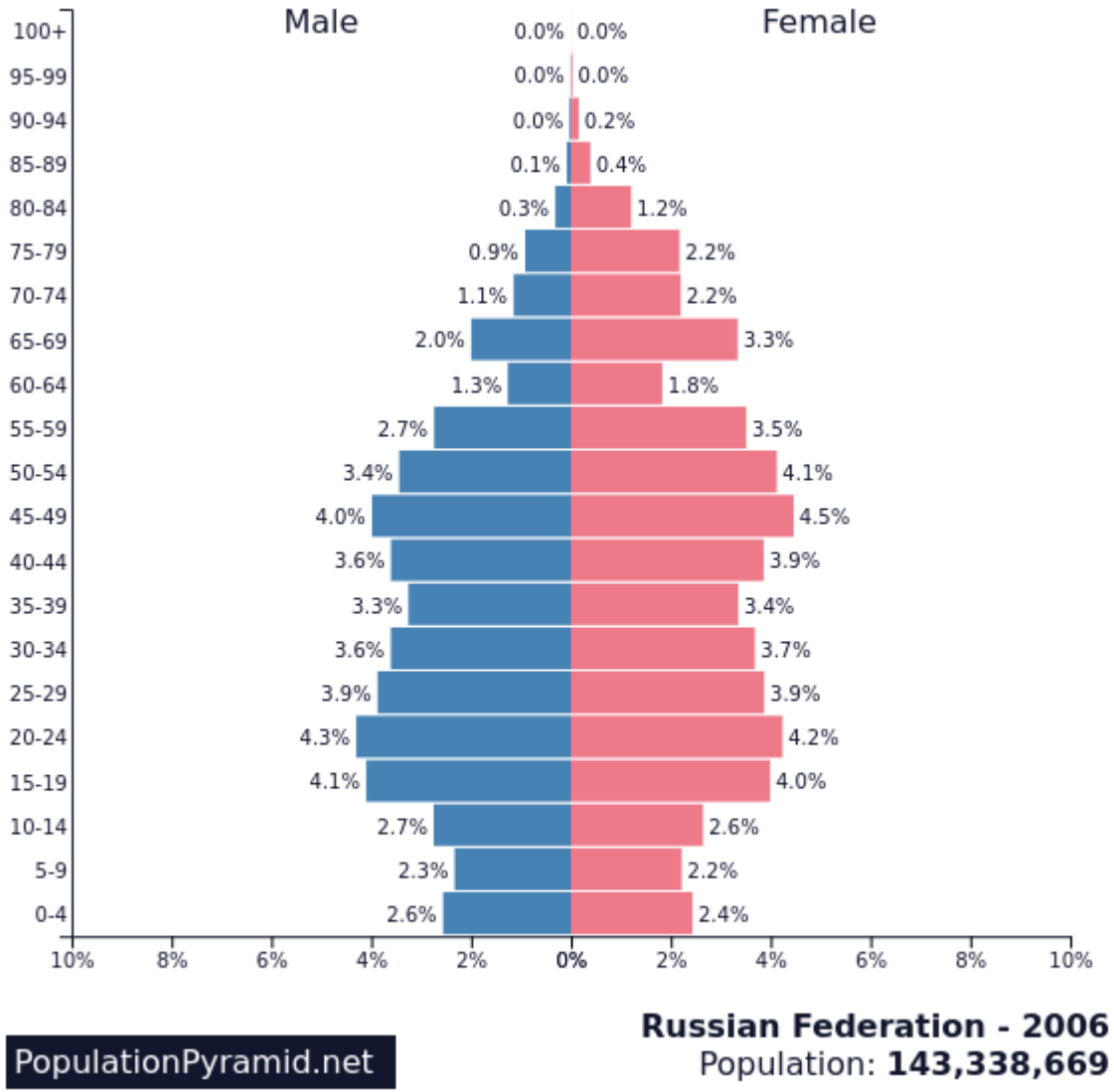

Male
Female
General fertility rate =
births in period
women aged 15-49
Reproductive Age
(15-49)
Age
100 +
95-99
90-94
85-89
80-84
75-79
70-74
65-69
60-64
55-59
50-54
45-49
40-44
35-39
30-34
25-29
20-24
15-19
10-14
5-9
0-4
15M 12.5M 10M 7.5M 5M 2.5M 0 2.5M 5M 7.5M 10M 12.5M 15M
Population
U.S. Census Bureau, International Database

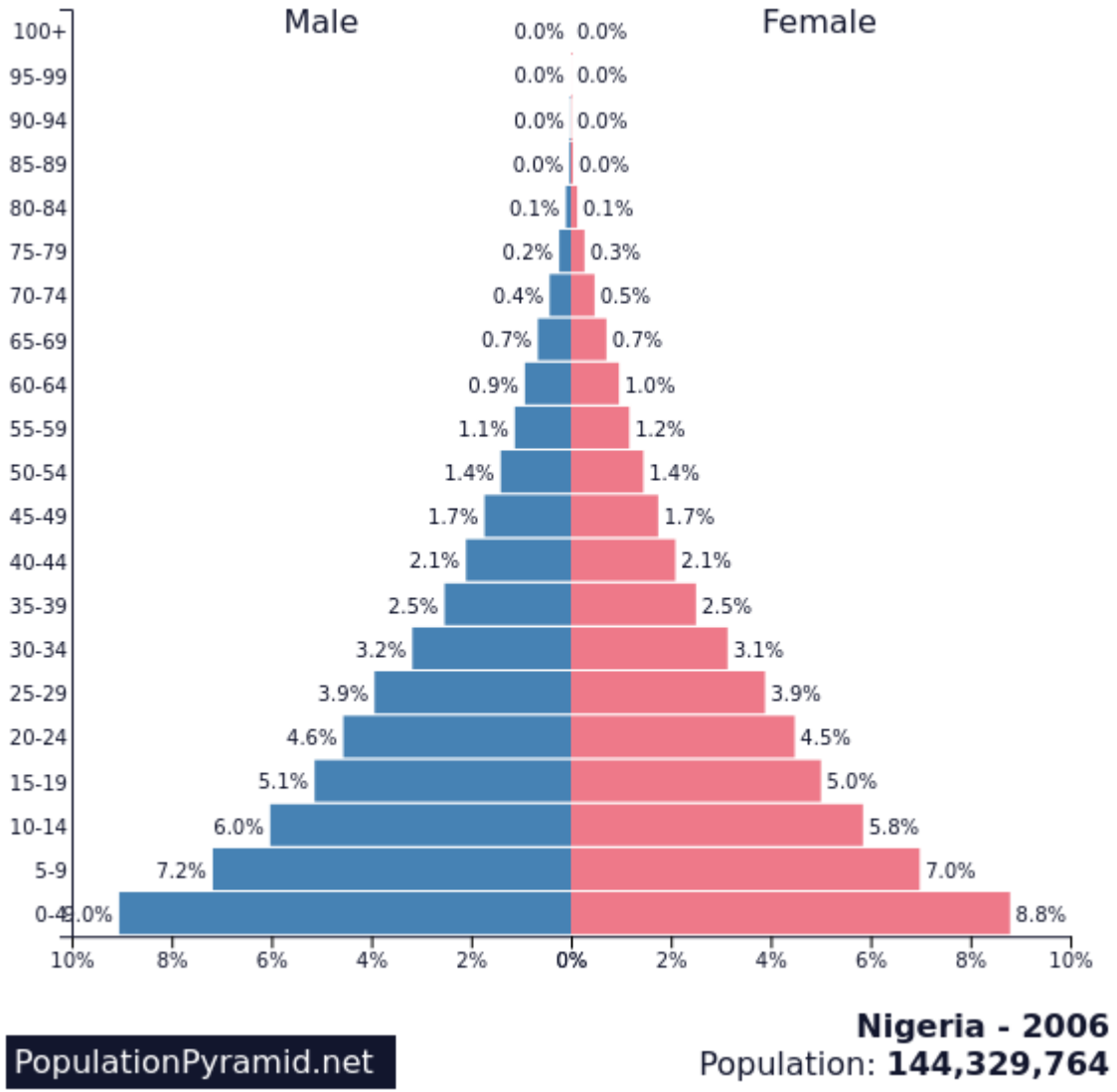
What do pyramids tell us about fertility rates?



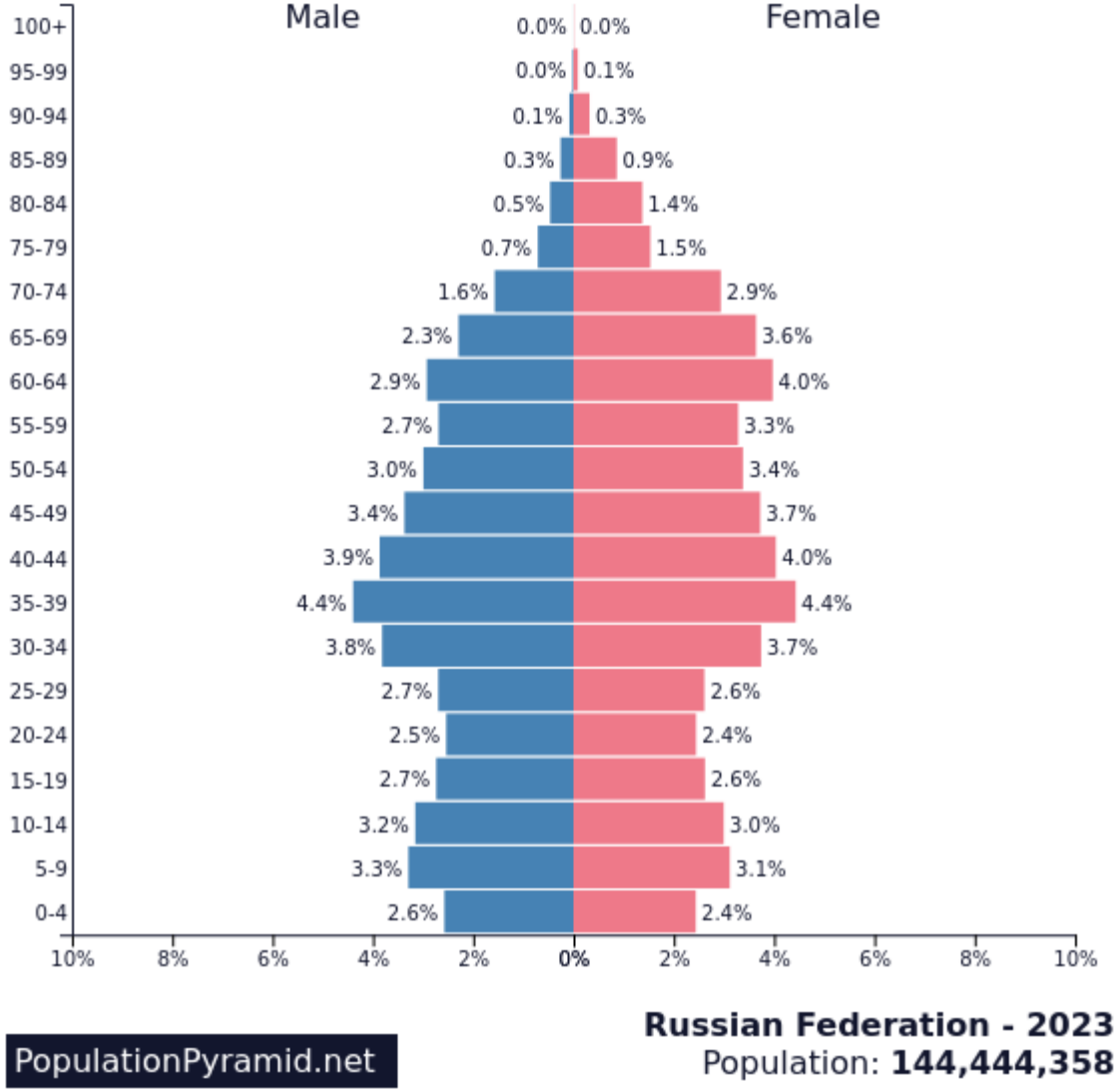


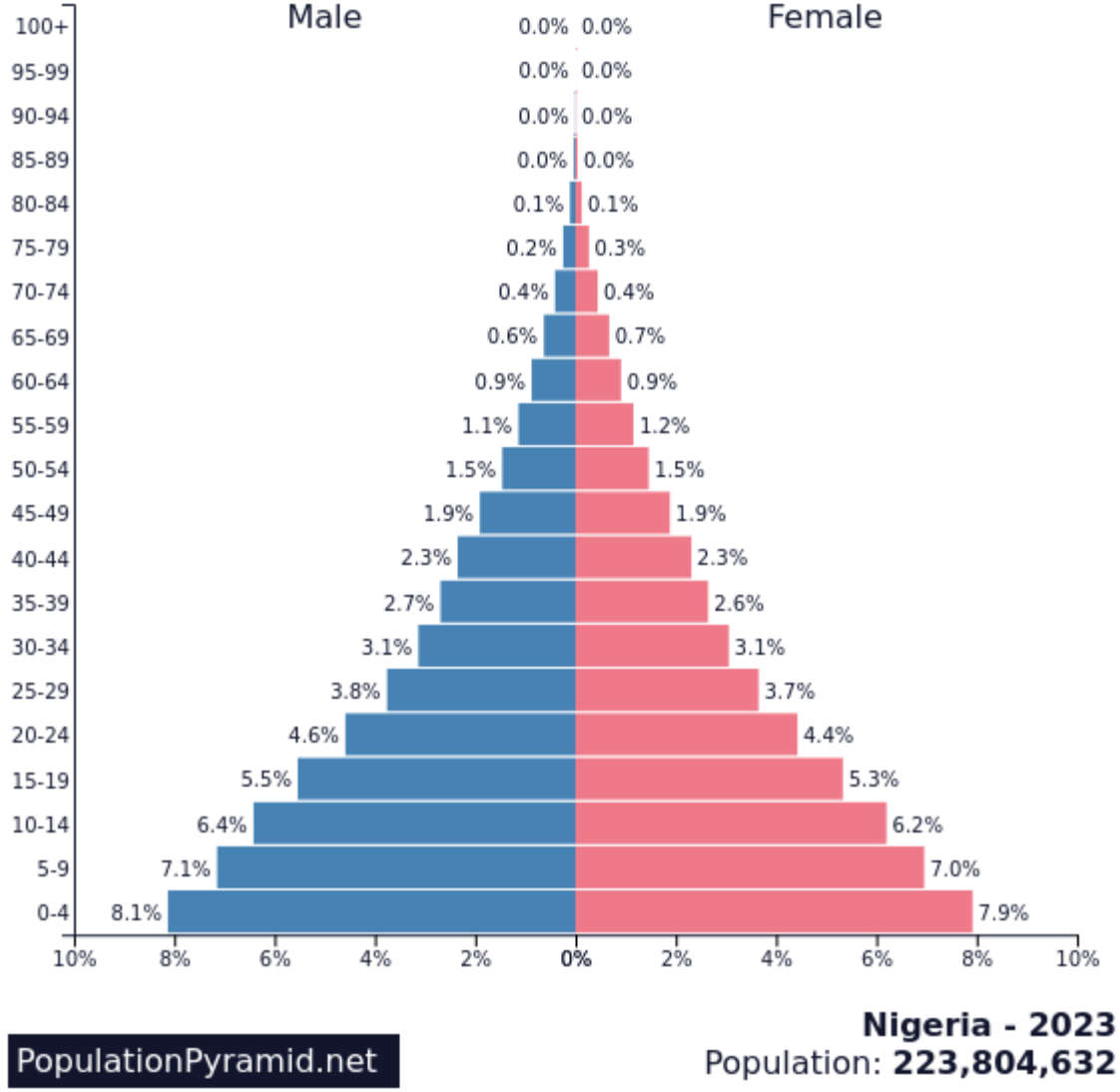
Both have a **crude death rate** of 16 deaths per 1,000 people





What do they look like now?





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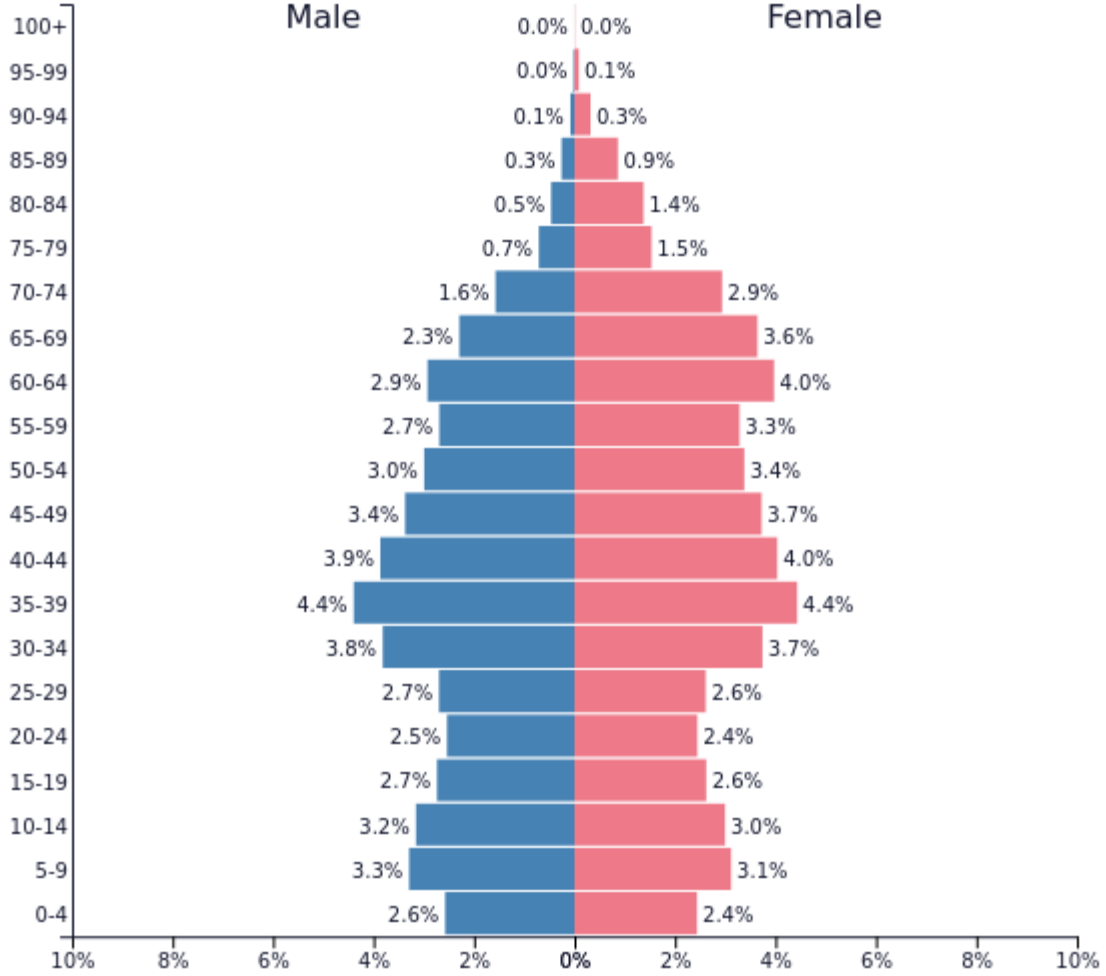
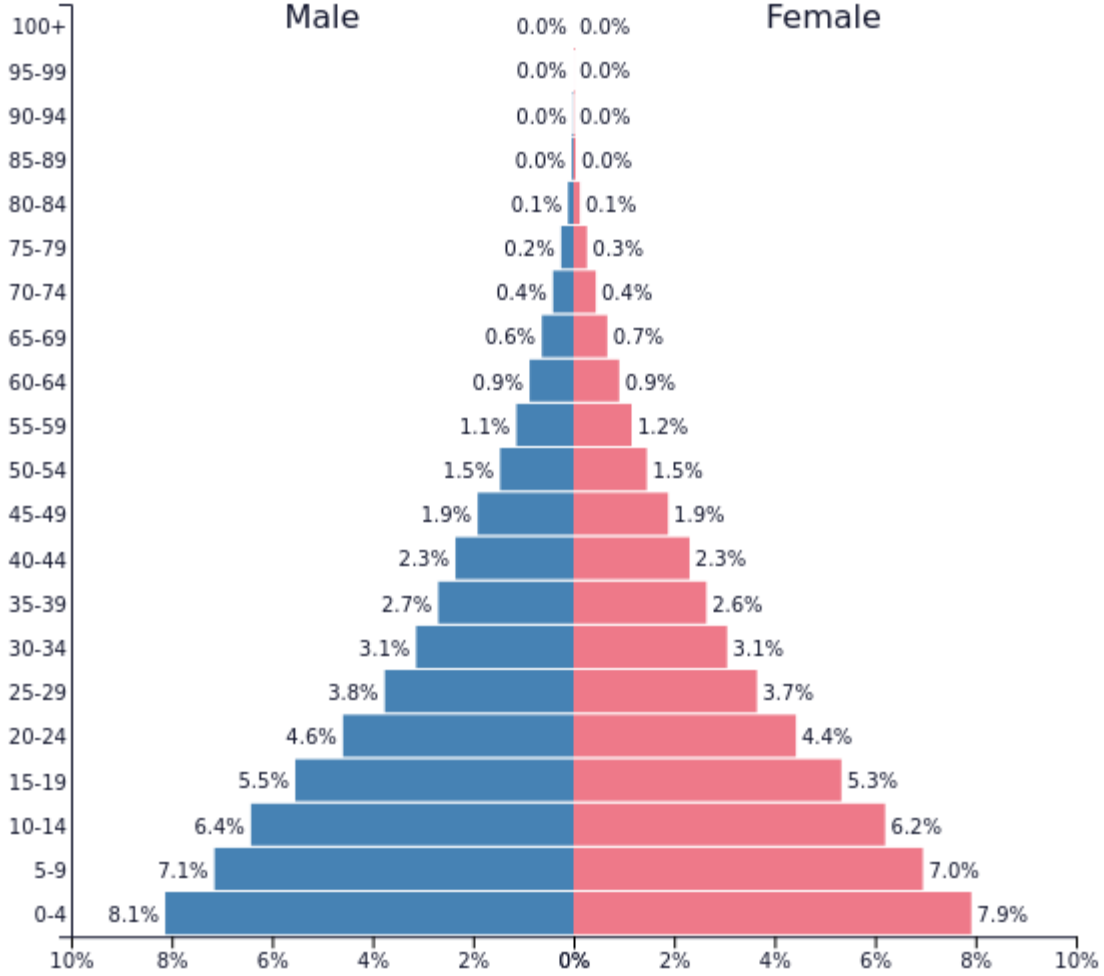


PopulationPyramid.net

Russian Federation - 2023
Population: **144,444,358**



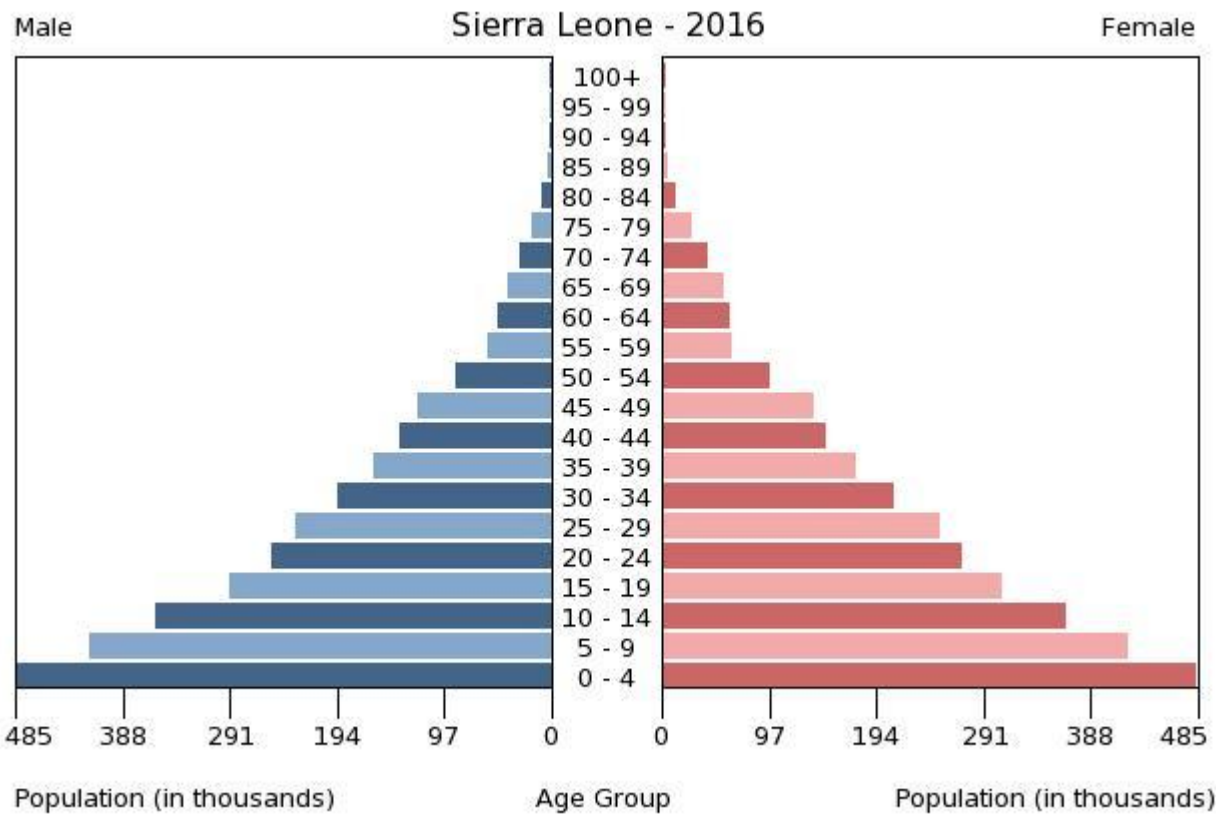
PopulationPyramid.net

Nigeria - 2023
Population: **223,804,632**

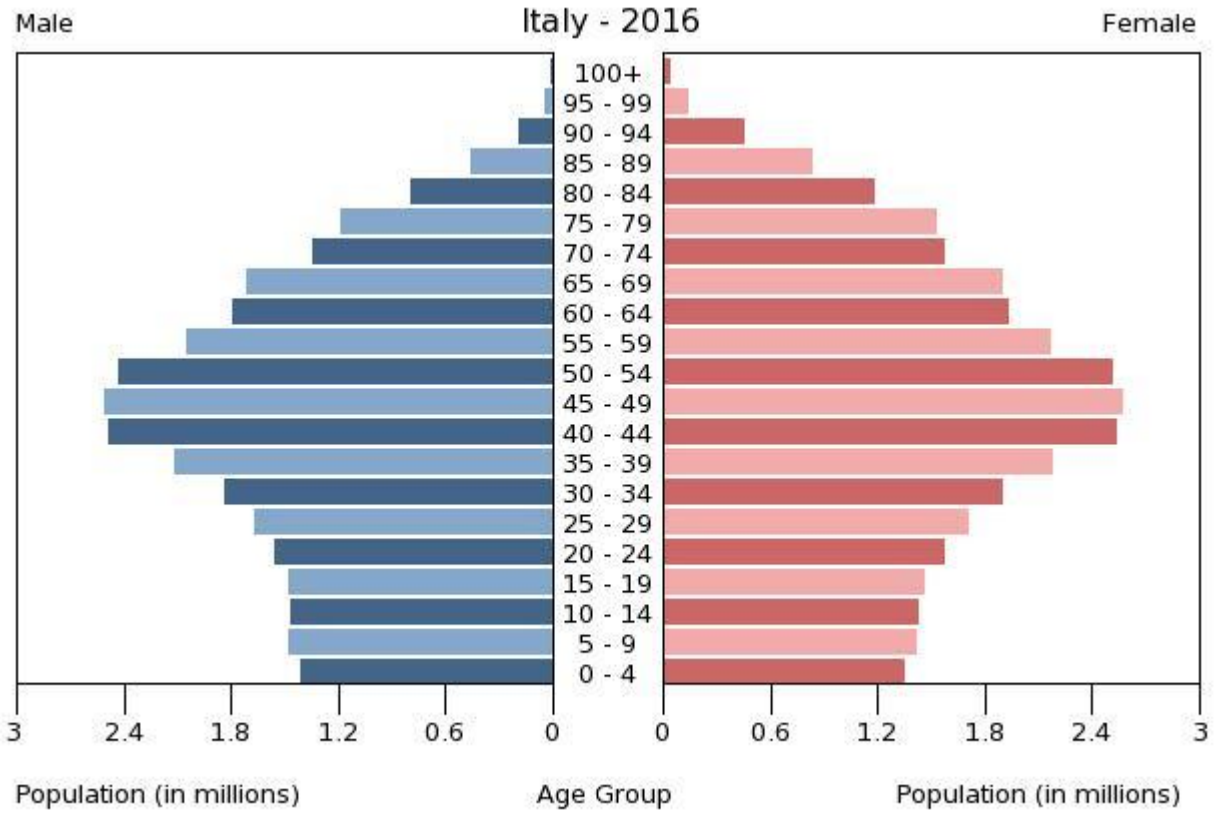
Why does age structure matter?

1. Record of past cohort experiences
2. Age pattern of demographic risk
3. Potential for population growth
4. Age dependency ratios

Age Structure and Population Growth Rates

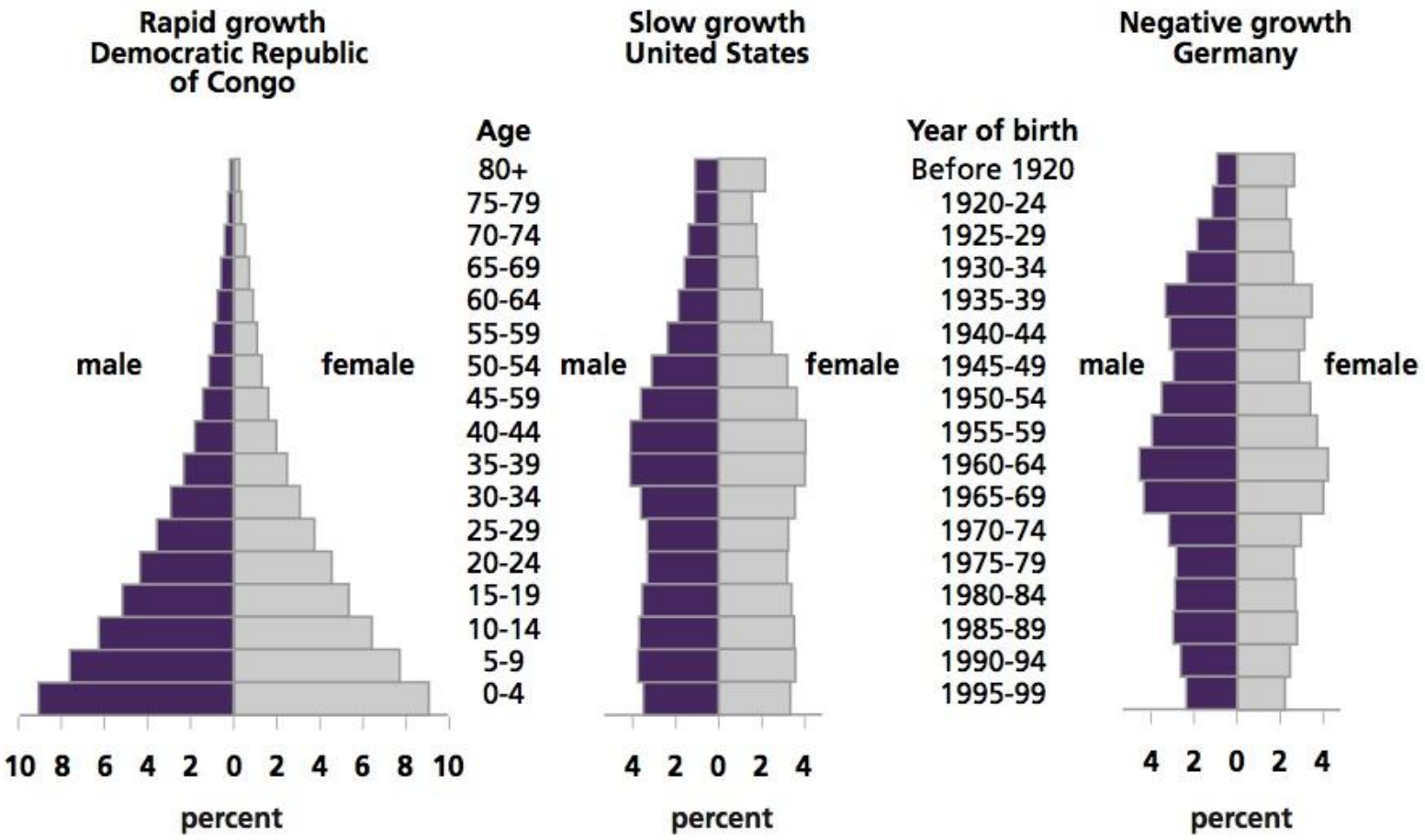


High fertility, high mortality
Growing population



Low fertility, low mortality
Shrinking population

Three patterns of population change, 2000



Source: United Nations, *World Population Prospects, The 1998 Revision*.

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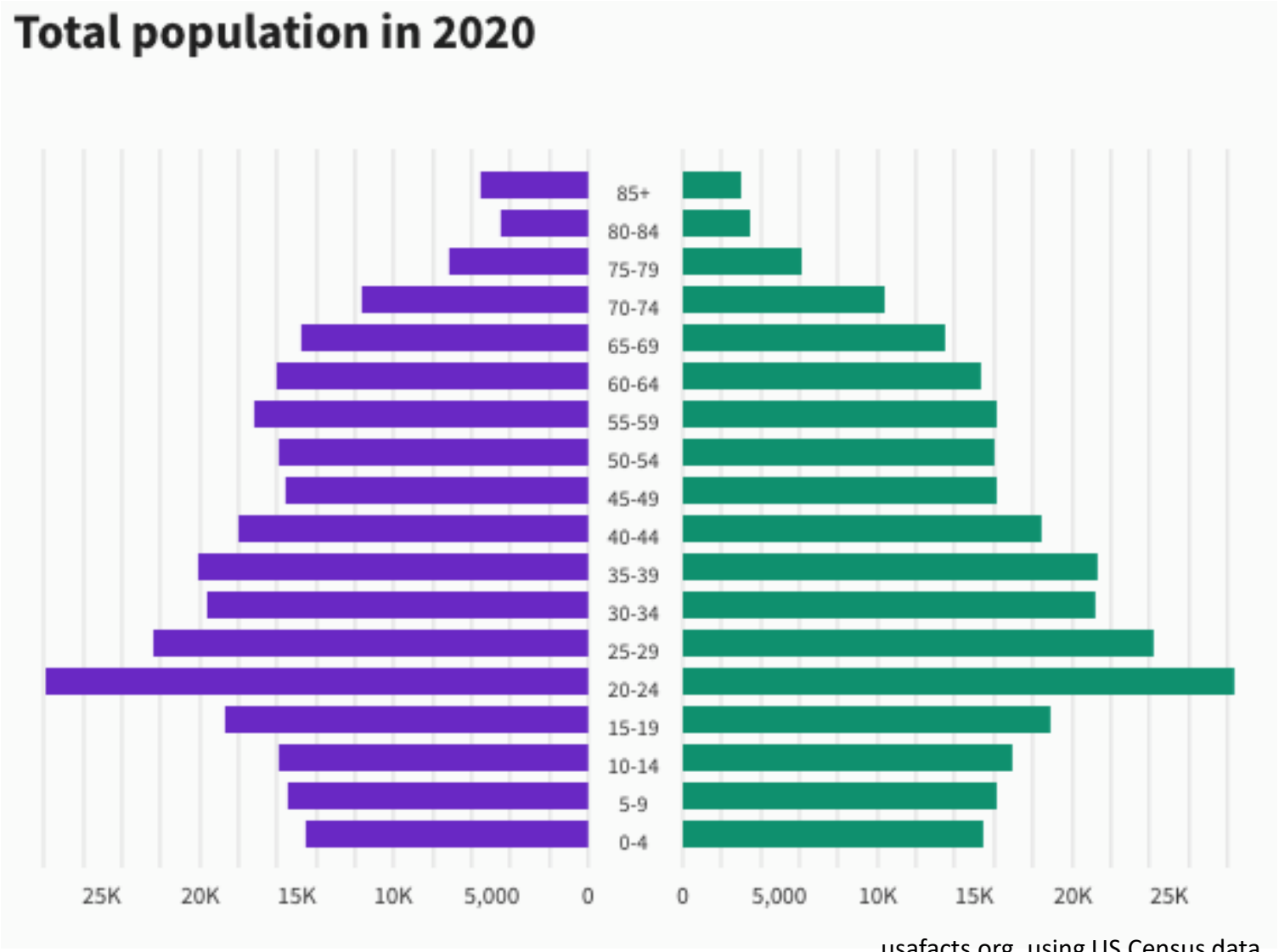
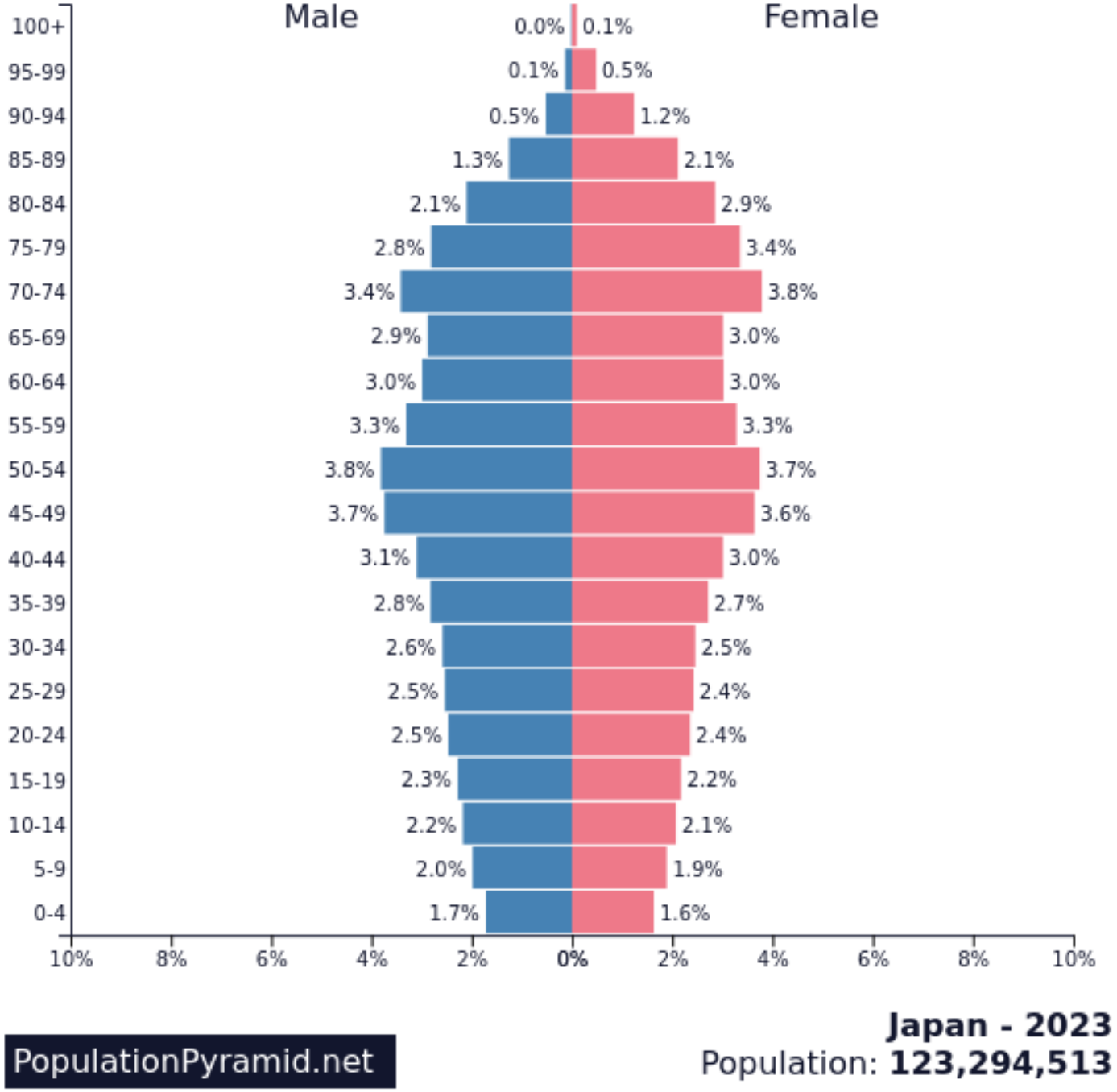
Dane County



usafacts.org, using US Census data

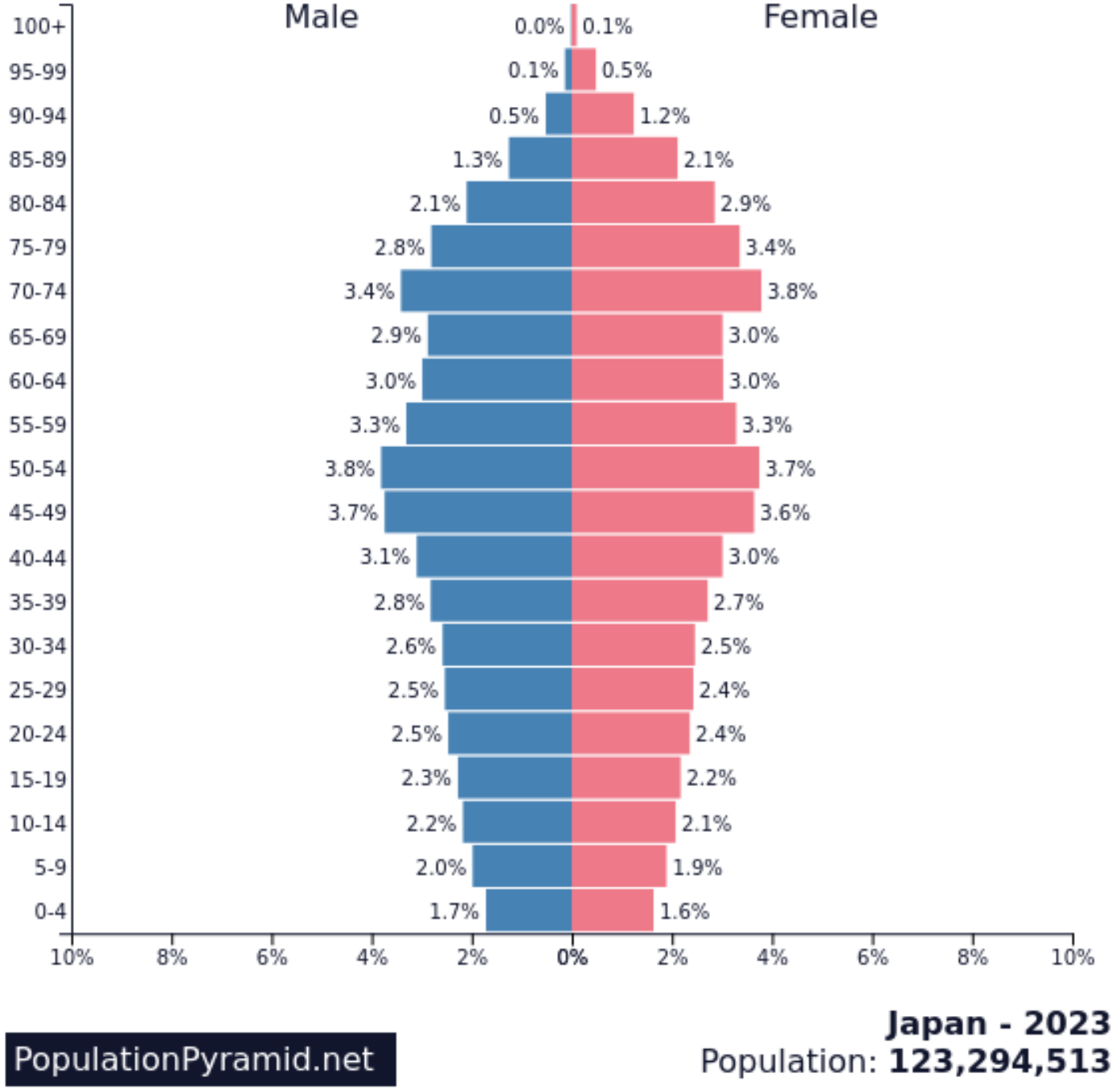
Predictions?



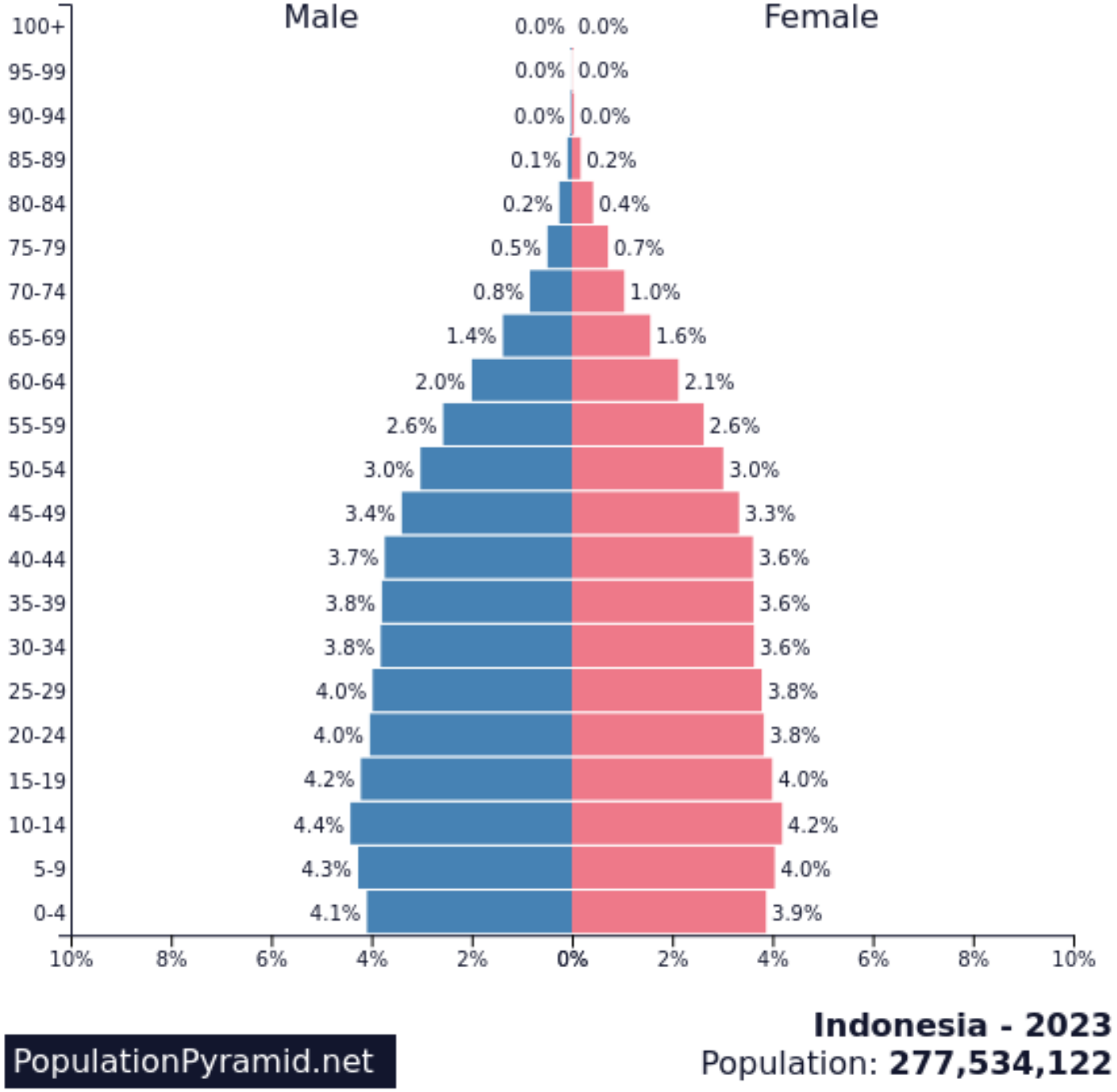
Predictions?

Projected population in 2050:
104,140,000

Decline by ~20 million (about 16%)



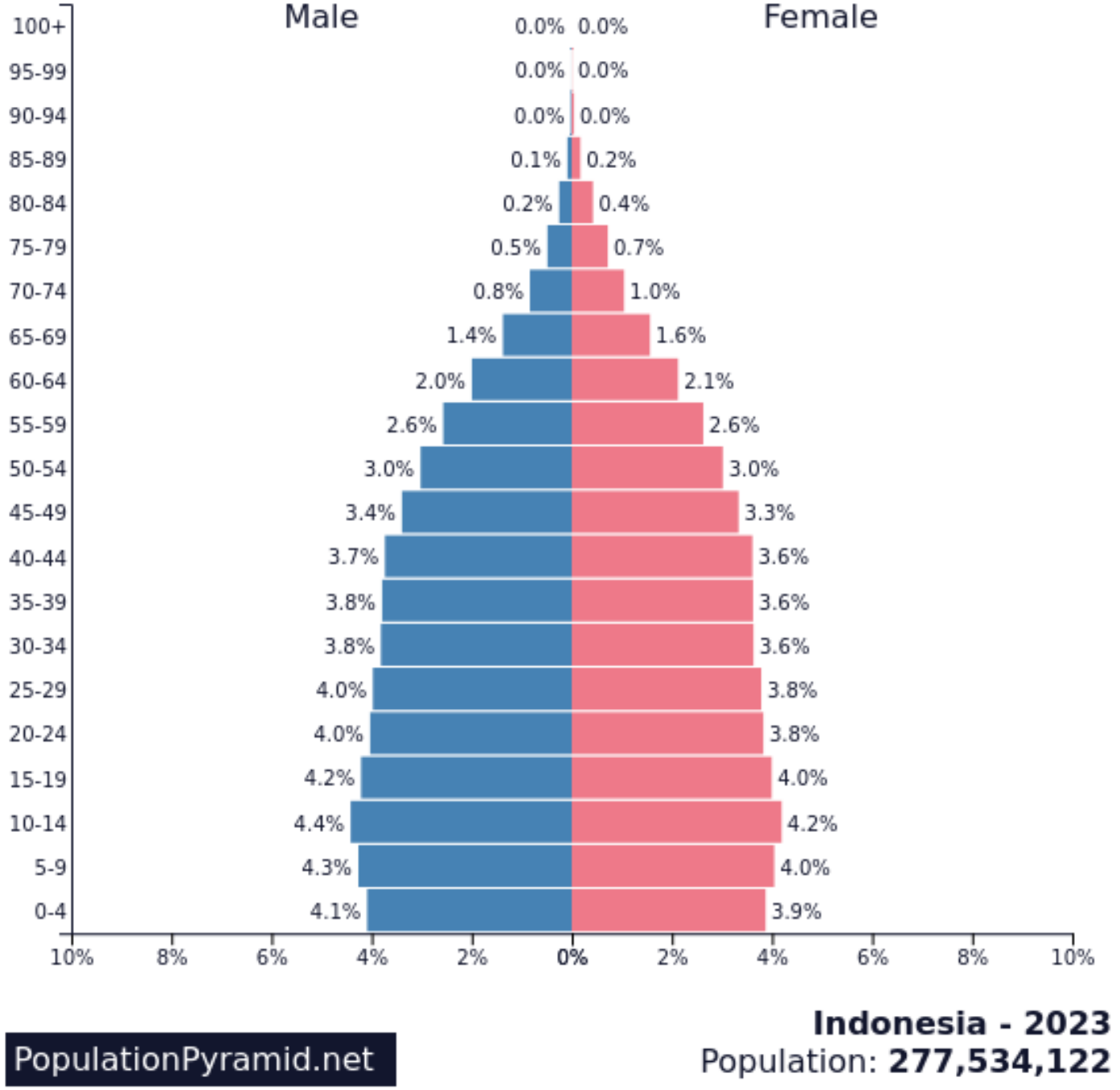
Predictions?



Predictions?

Projected population in 2050:
316,968,000

Increase by ~40 million (about 14%)

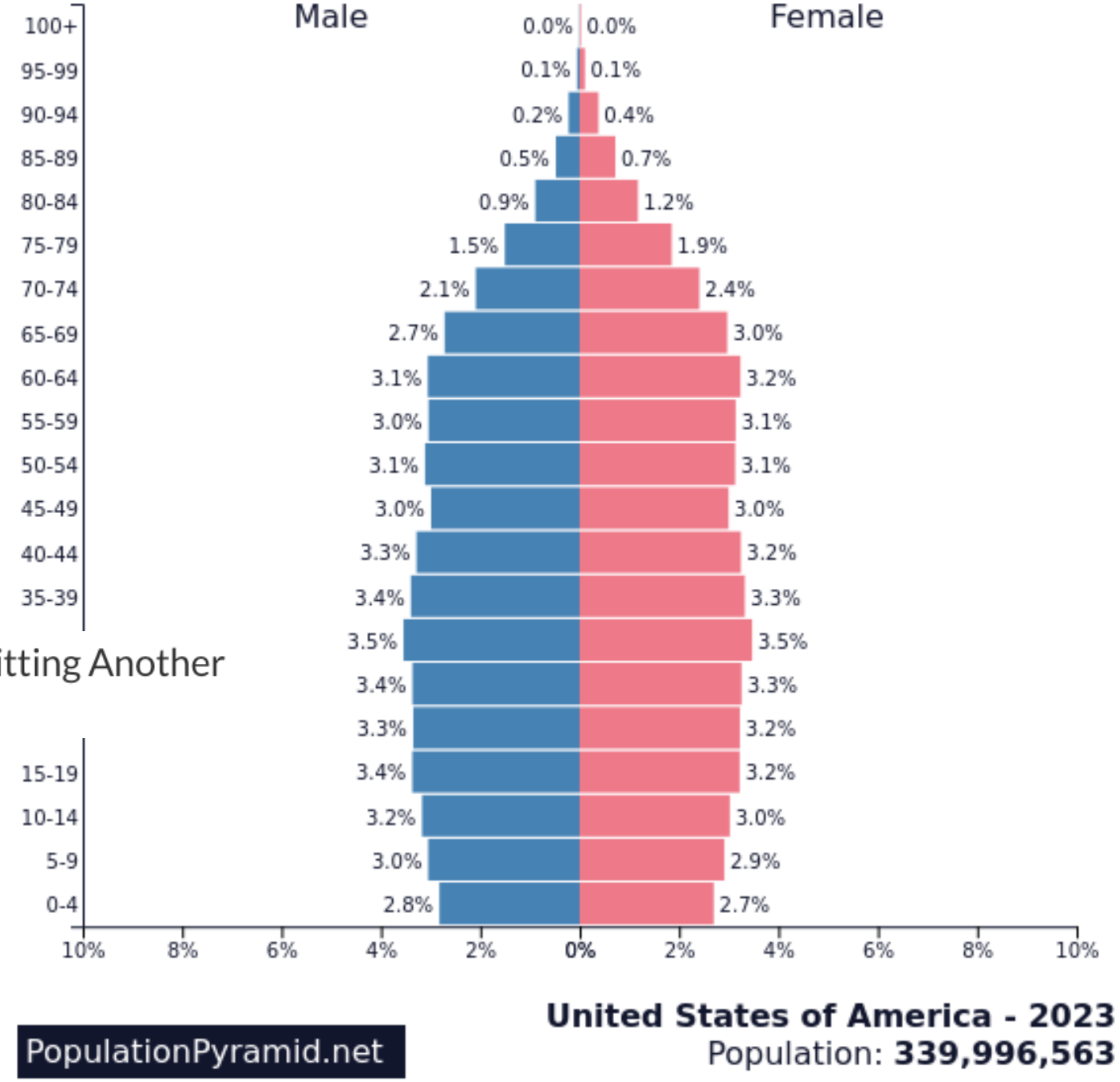


Predictions?

The U.S. Birthrate Has Dropped Again. The Pandemic May Be Accelerating the Decline.

US birth rate falls to lowest point in more than a century

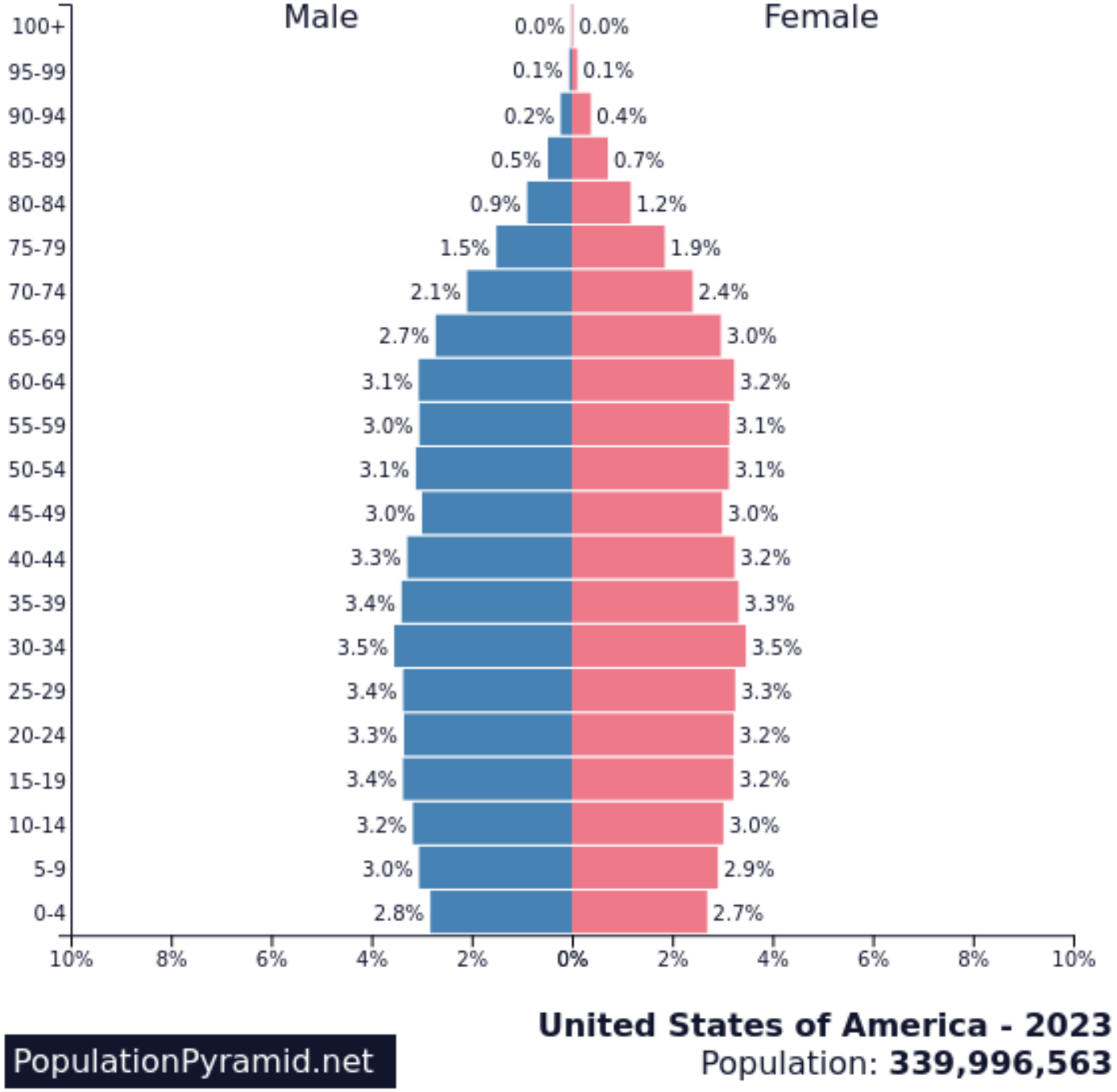
U.S. Birthrate Fell By 4% In 2020, Hitting Another Record Low



Predictions?

Projected population in 2050:
375,085,000

Increase by ~35 million (about 10%)



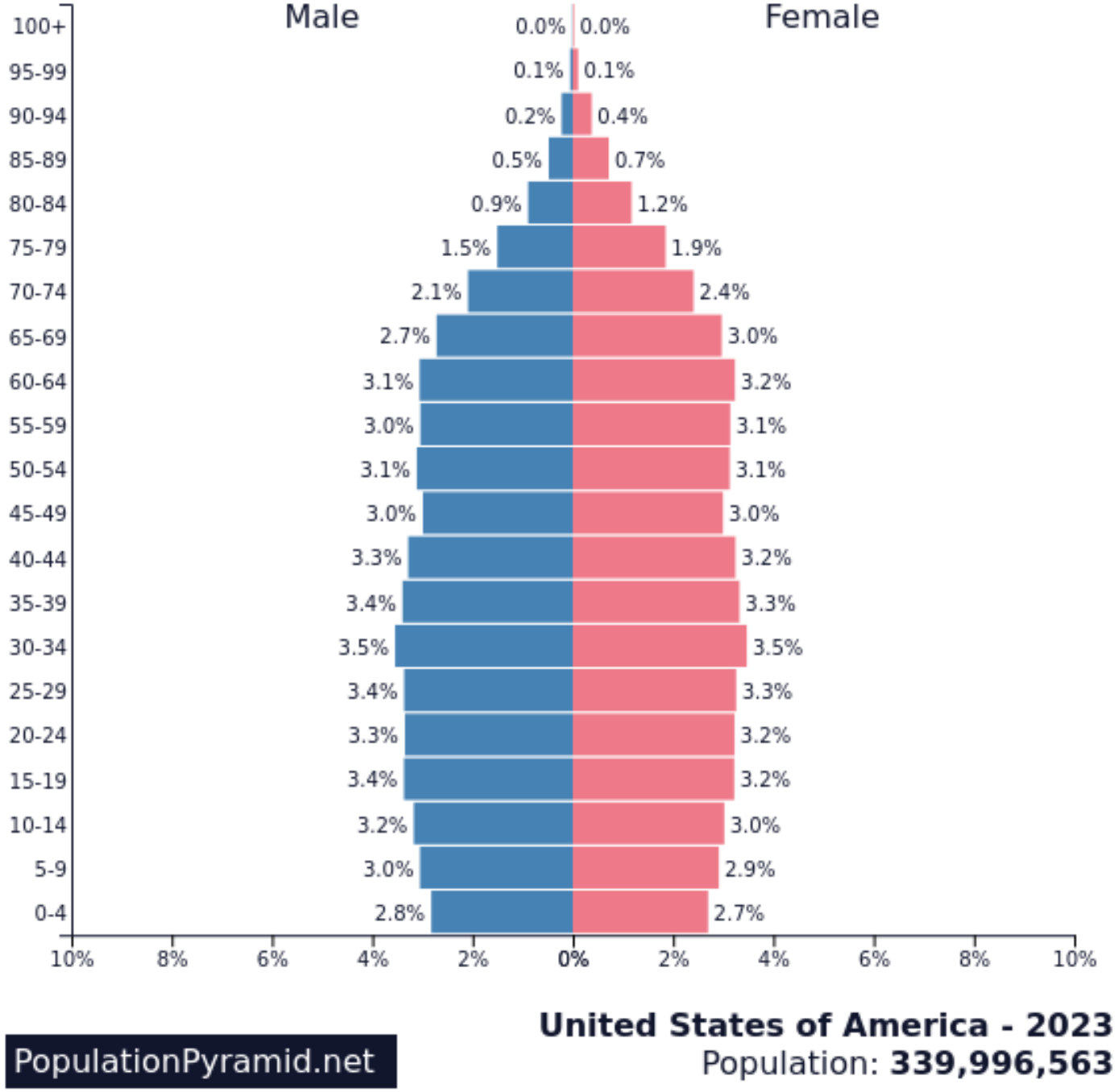
Predictions?

Projected population in 2050:
375,085,000

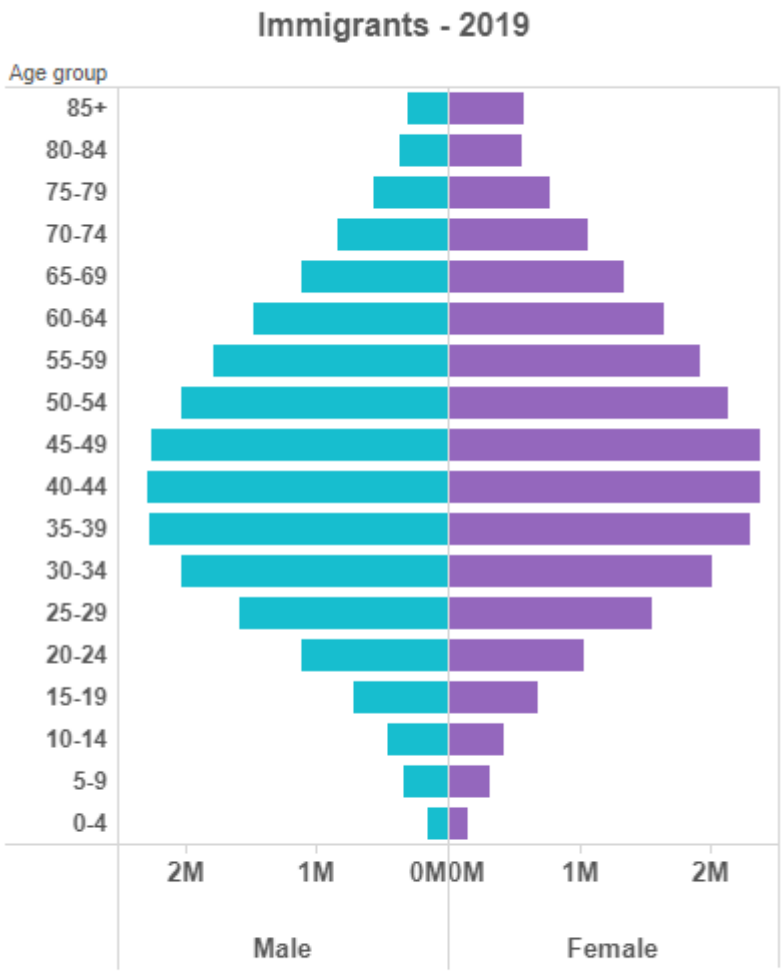
Increase by ~35 million (about 10%)

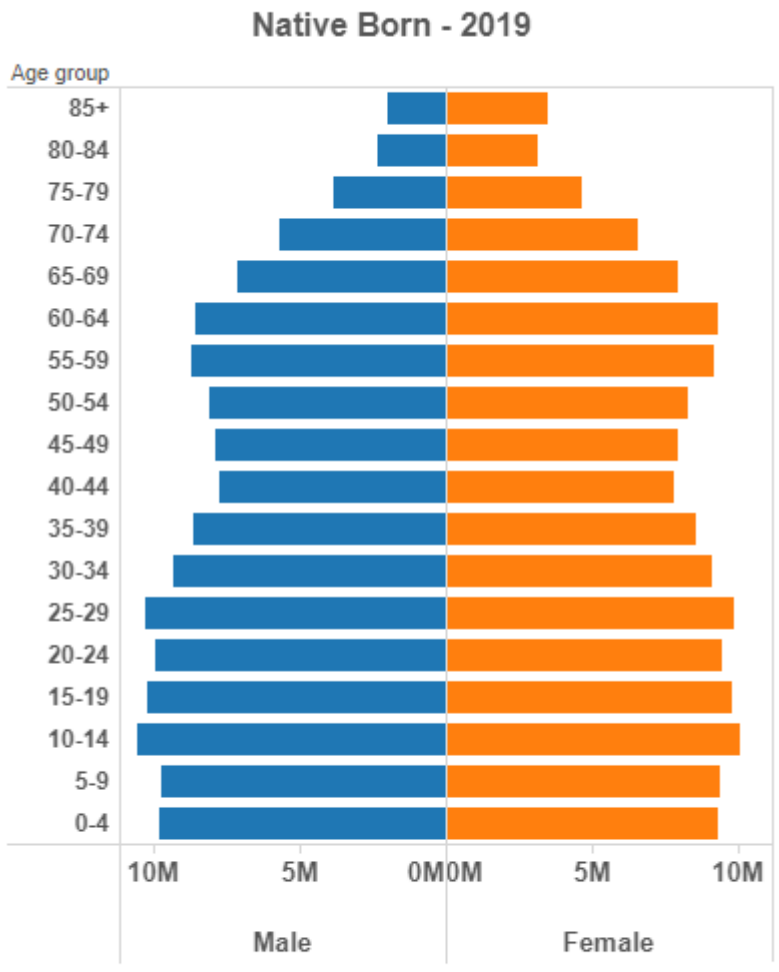
2 reasons:

1) **Population Momentum**
   - Population continues to increase even though fertility declines
   - Baby boomers had a lot of kids too! (Echo)
2) **Immigration**



Age and Sex Distribution of the U.S. Immigrant and Native-Born Populations





Total U.S. immigrant population: 44,932,900

Total U.S. native population: 283,306,600

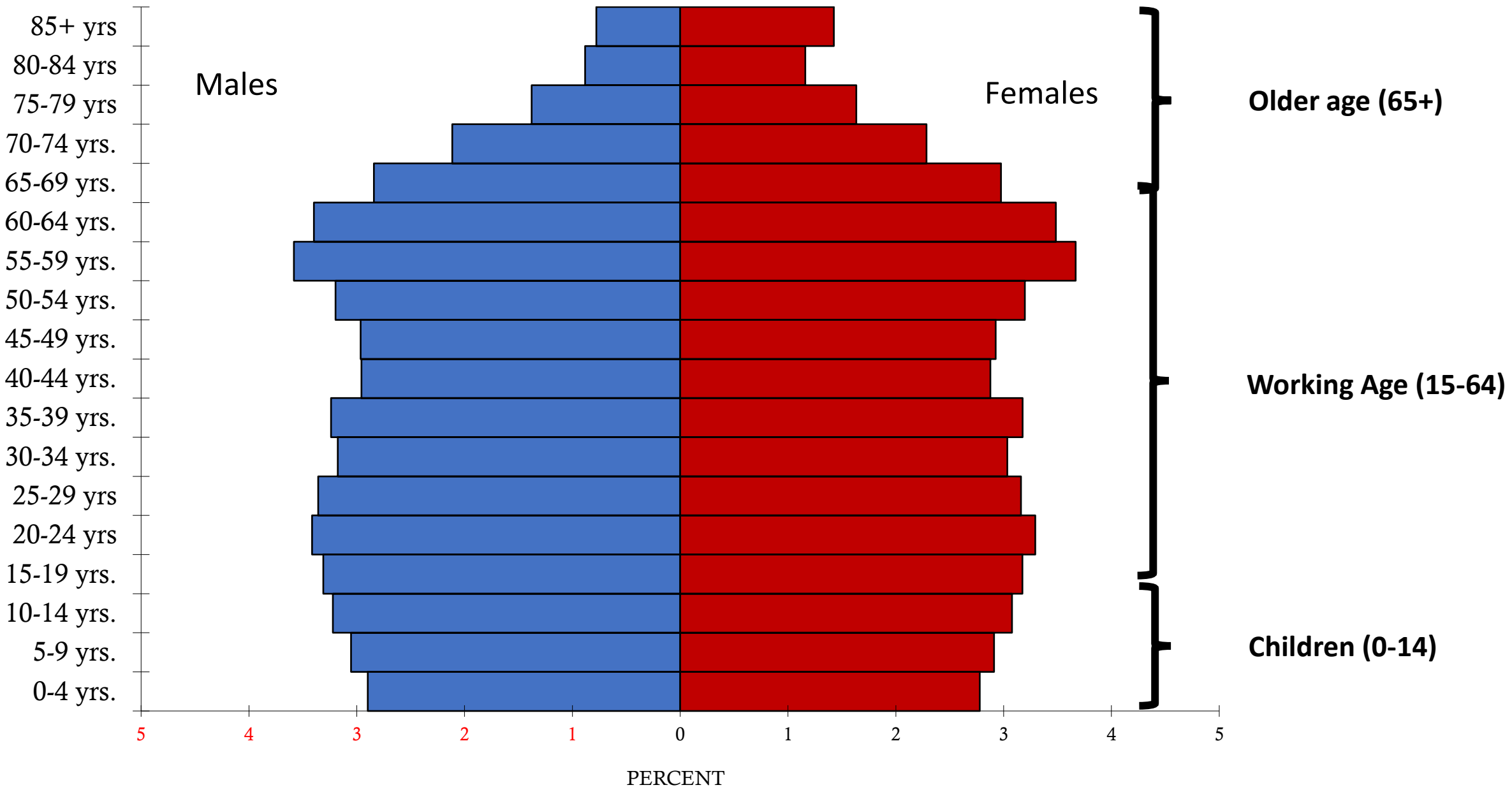
Migration Policy Institute (MPI) Data Hub
http://migrationpolicy.org/programs/data-hub

Year
2019

Why does age composition matter?

1. Record of past cohort experiences
2. Age pattern of demographic risk
3. Potential for population growth
4. Age dependency ratios

United States: Age-Sex Distribution, 2020 (Dependent Groups)



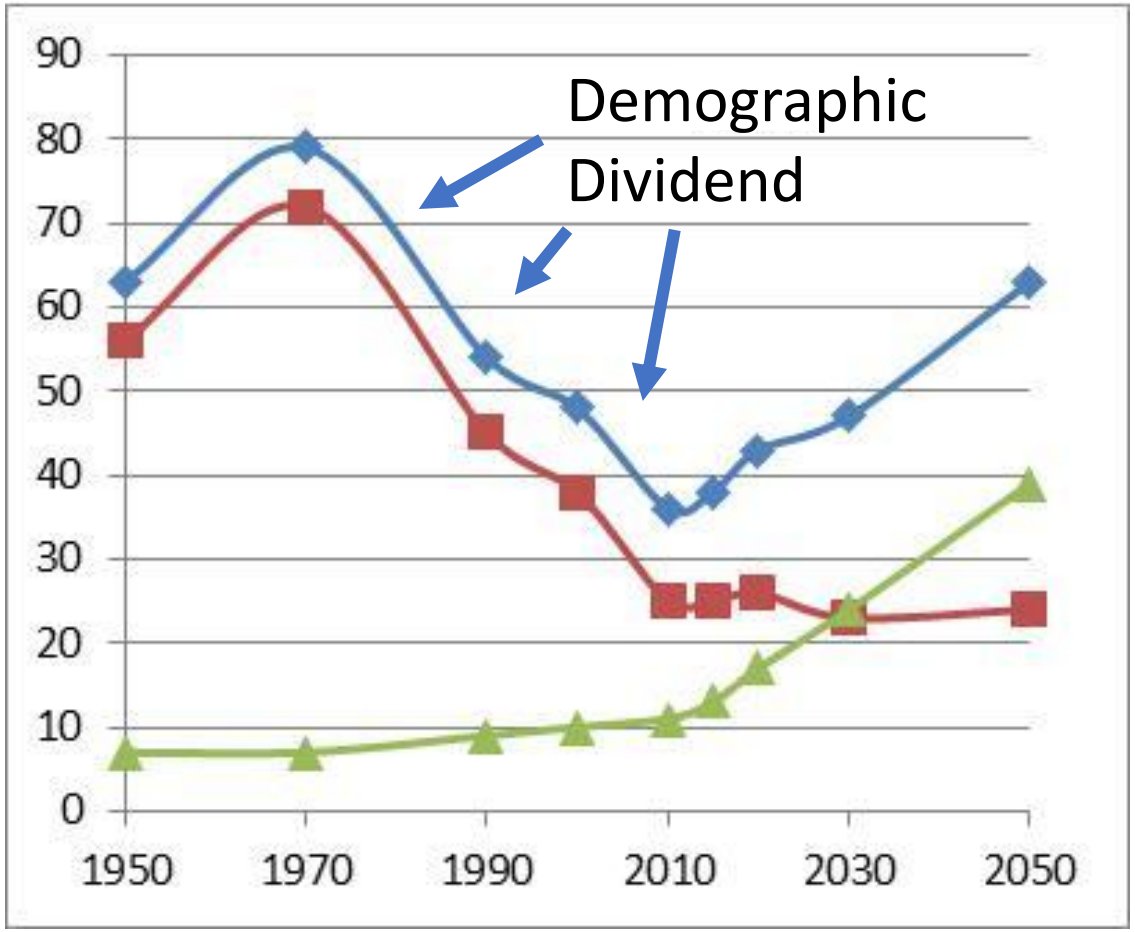
Old-Age Dependency Ratio

$$\frac{\#\text{ of old age dependents (people aged}>64)}{\#\text{ supporters (people aged } 15-64)}.$$

(or 18-64)
(or 20-64)

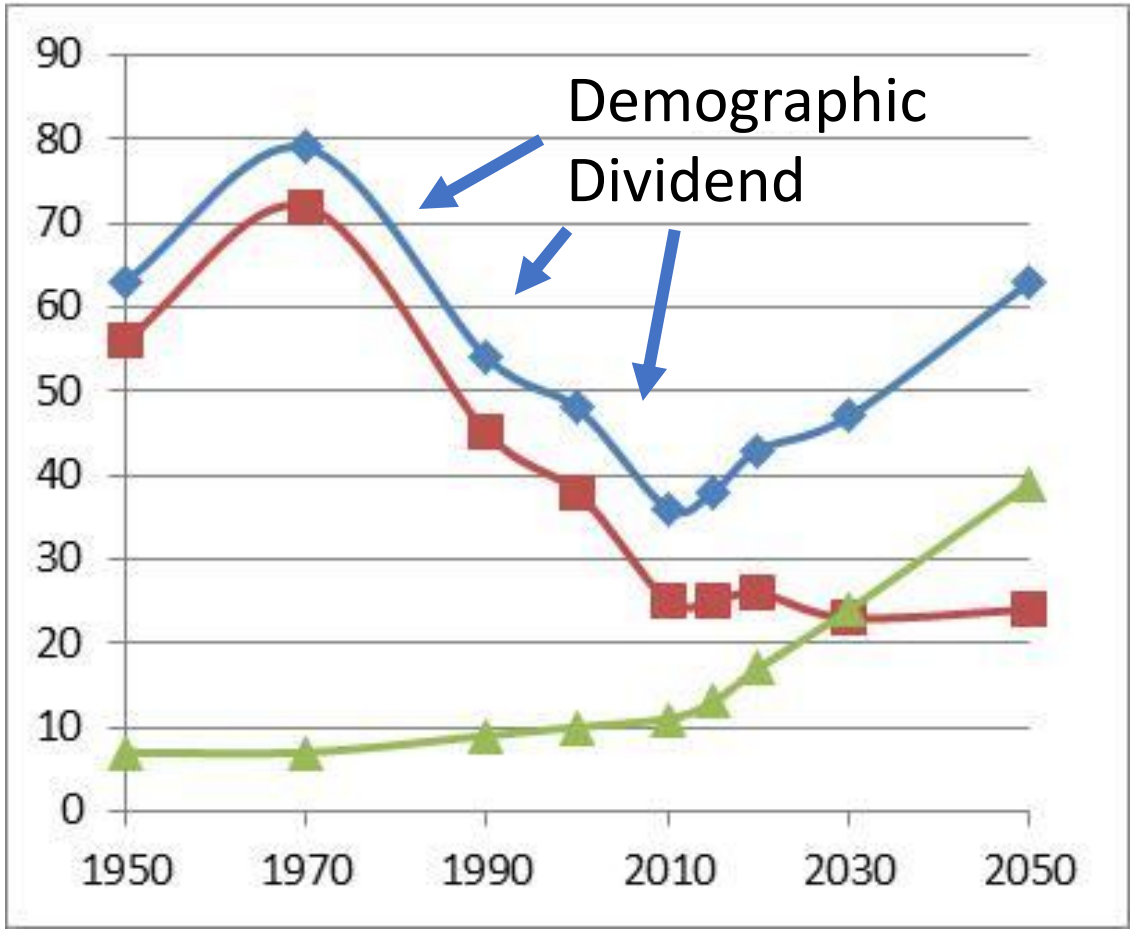
China had very high fertility, then a rapid decline, with huge implications for Dependency Ratios!

Dependency Ratios: China



Blue = Total DR
Red = Children DR
Green = Old Age DR

Dependency Ratios: China



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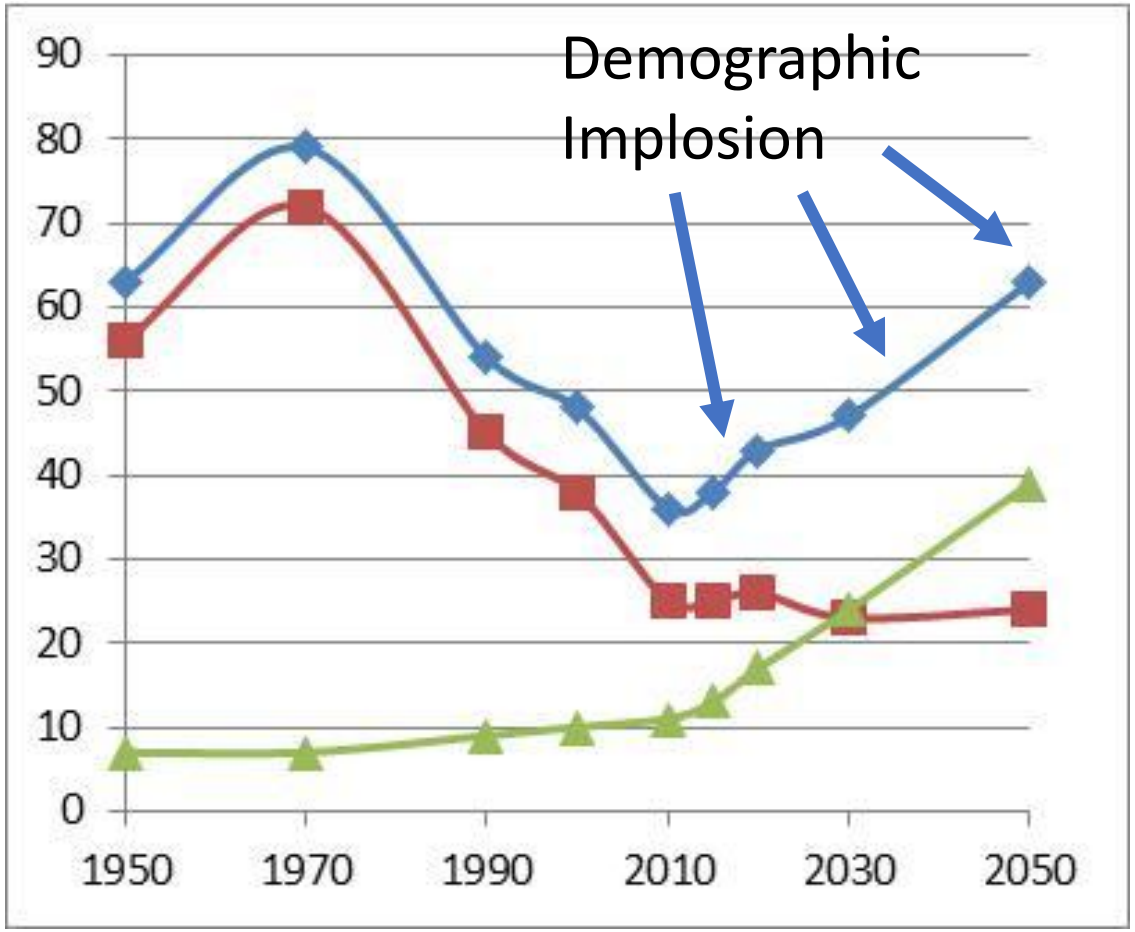
Demographic Dividend:

- Low dependency ratios mean high fraction of people working and spending money → economic growth

Dependency Ratios: China



Blue = Total DR
Red = Children DR
Green = Old Age DR

Demographic Implosion:
- A sharp increase in the dependency ratio

2010: 36 dependents per 100 working age adults

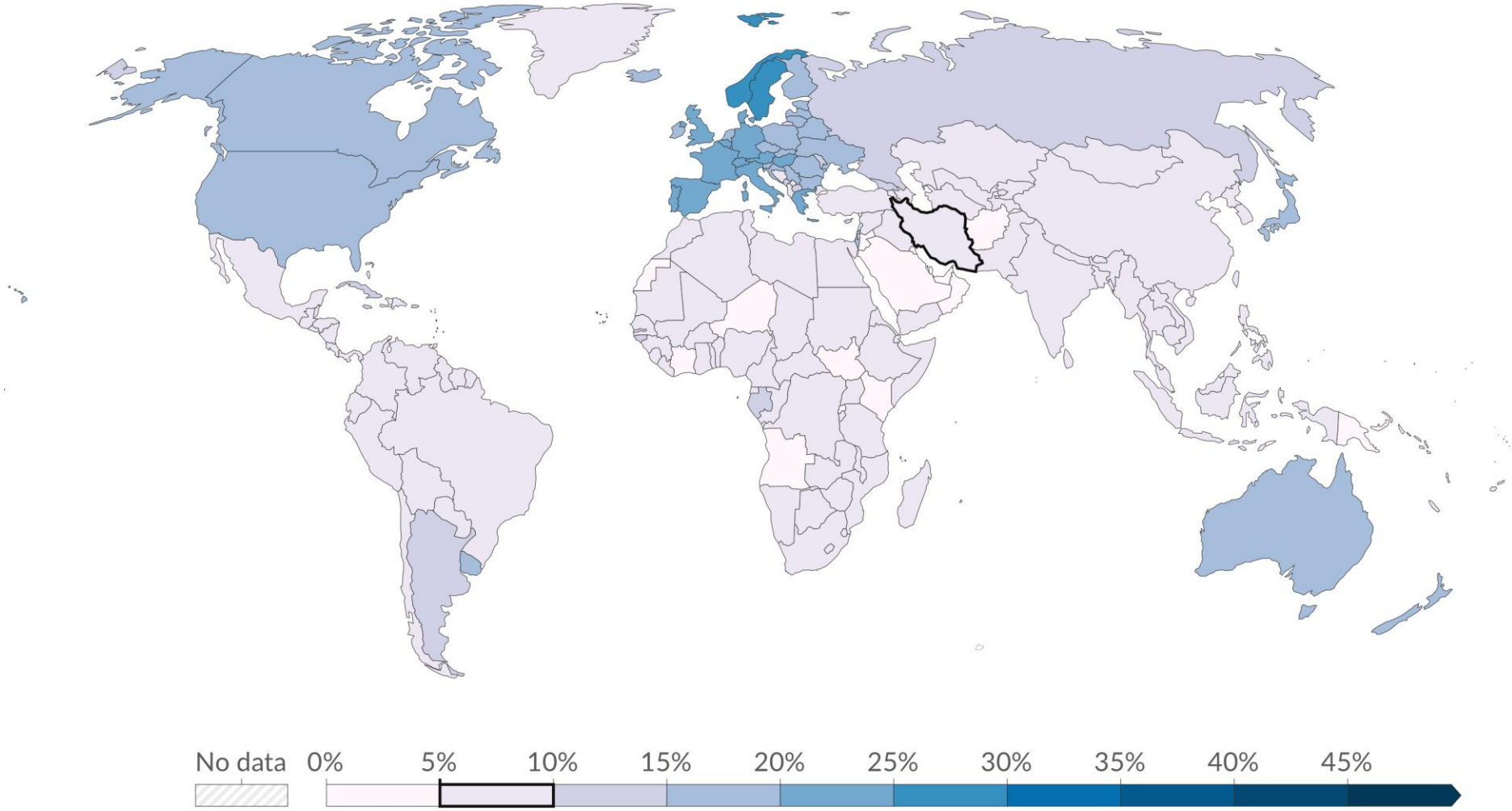
2030: 47

2050: 60+

Old-age dependency ratio, 1990

Our World in Data

The number of elderly people (over 64 years old) relative to the number of working-age people (15-64 years). Figures are shown as the number of dependents per 100 working-age people. A value of 20% means that there are 20 people of old age for every 100 working-age people.



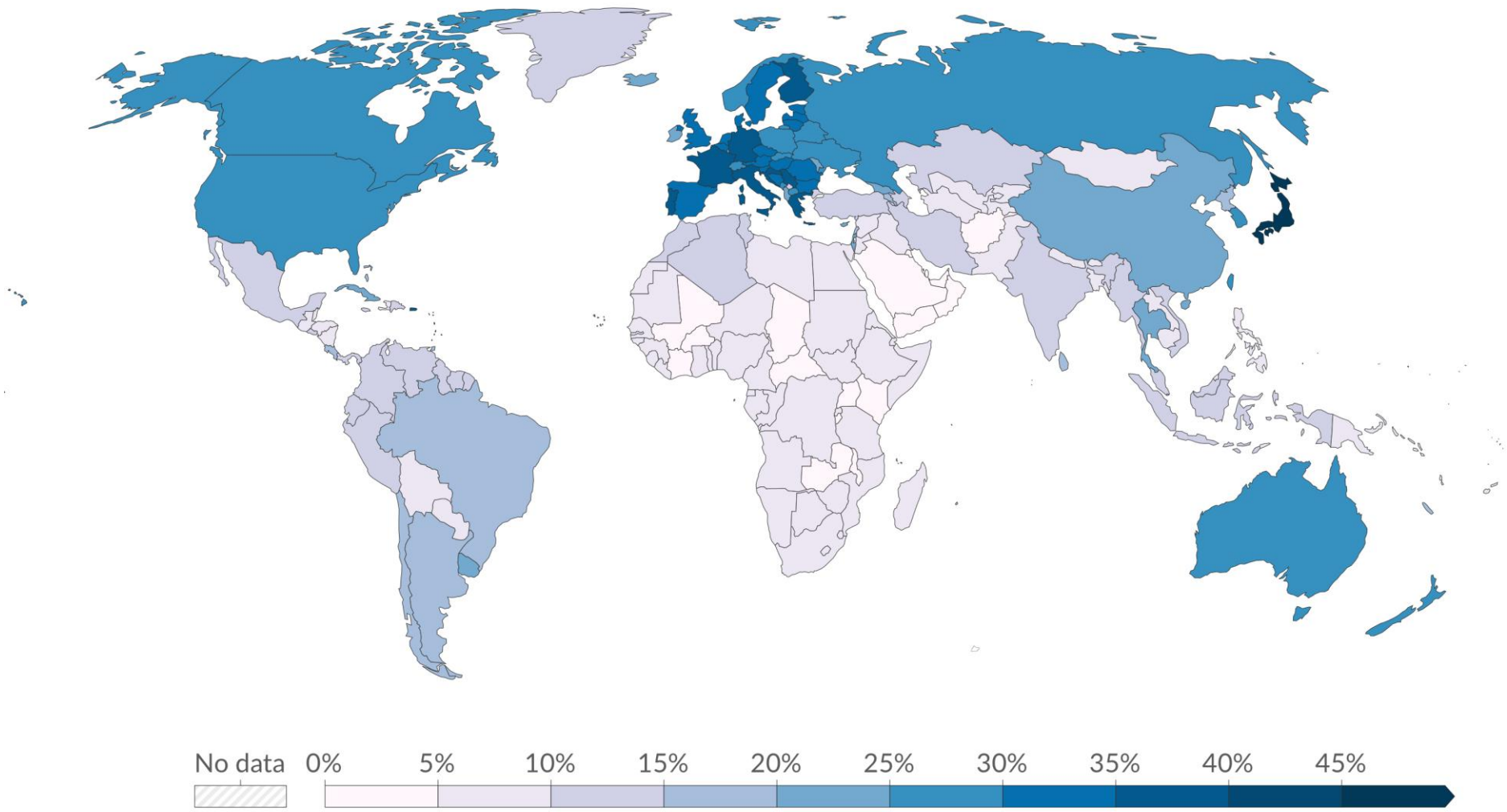
Data source: UN, World Population Prospects (2024)

OurWorldinData.org/age-structure | CC BY

Old-age dependency ratio, 2023

Our World in Data

The number of elderly people (over 64 years old) relative to the number of working-age people (15-64 years). Figures are shown as the number of dependents per 100 working-age people. A value of 20% means that there are 20 people of old age for every 100 working-age people.



Data source: UN, World Population Prospects (2024)

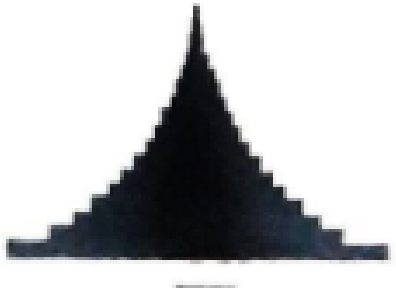
OurWorldinData.org/age-structure | CC BY

Question of the day

https://tinyurl.com/170Jan22

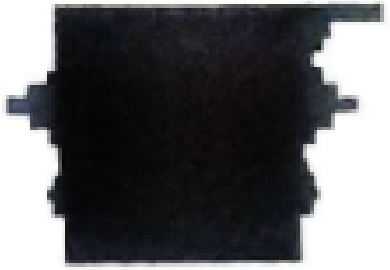
Today we discussed why age composition matters for societies. Which of the four reasons do you think is most important, and why? Answer in 1-2 sentences.

Types of Population Pyramids



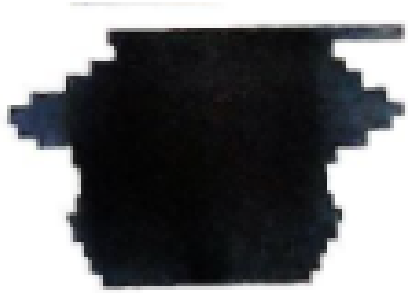
Young

Mature



Old

Undercut



Declining

Unimodal